



Design and Simulation of a Light Activated Relay Power Driver with Electrical Isolation and Flyback Protection

Academic electronics design and Proteus simulation report

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This report presents the design, calculation basis, circuit scheme, simulation procedure, and technical explanation of a light activated relay power driver. The system uses a low voltage sensing circuit to control a relay, while the relay contact side is used to switch a higher power load. The design includes a light dependent resistor, a comparator based control stage, a relay driver stage, a relay coil, relay contacts, and a flyback diode for inductive voltage suppression.

Abstract

This report investigates the design and Proteus simulation of a light activated relay power driver intended to control a higher power load using a low voltage sensing circuit. The main purpose of the project is to demonstrate how a low power electronic control stage can operate a relay coil, while the relay contacts switch a separate load circuit. The system uses a light dependent resistor as the sensing element, a voltage divider to convert light intensity into a variable voltage, a reference voltage to define the switching threshold, a comparator stage to determine whether the environment is bright or dark, and a driver stage to supply the relay coil current.

The relay provides functional separation between the control circuit and the load circuit. When the light level decreases below the selected threshold, the comparator output changes state, the driver transistor or MOSFET conducts, and the relay coil becomes energized. As a result, the normally open relay contact closes and the load is activated. A flyback diode is connected across the relay coil to suppress the reverse voltage produced when the coil current is interrupted.

The report explains the operating principle of each stage, the mathematical relationships used in the design, and the expected behavior of the complete system. It also emphasizes that relay based switching must include correct coil driving, correct diode polarity, correct contact selection, and clear separation between the low voltage control side and the higher power load side.

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Symbols and Abbreviations

LDR means light dependent resistor. It is a resistor whose resistance changes according to incident light intensity.

VCC means the positive direct current supply voltage of the control circuit. In this project, VCC is normally taken as 5 volts.

GND means the zero volt reference point of the low voltage control circuit.

VSENSE means the voltage produced by the LDR voltage divider. This voltage represents the detected light condition.

VREF means the reference voltage used by the comparator. It defines the boundary between the bright condition and the dark condition.

VOUT means the output voltage of the comparator stage.

ICOIL means the current flowing through the relay coil when the relay is energized.

RCOIL means the electrical resistance of the relay coil.

COM means common relay contact.

NO means normally open relay contact. This contact is open when the relay coil is not energized and closes when the relay coil is energized.

NC means normally closed relay contact. This contact is closed when the relay coil is not energized and opens when the relay coil is energized.

AC means alternating current.

DC means direct current.

SPDT means single pole double throw relay contact arrangement.

Flyback diode means a diode connected across an inductive coil to provide a safe current path when the coil is switched off.

Component List Summary

The project requires a 5 volt direct current source for the control circuit, a light dependent resistor for light sensing, a fixed resistor for the LDR voltage divider, a potentiometer or resistor divider for generating the reference voltage, a comparator for threshold detection, a pull up resistor for the comparator output when an open collector comparator is used, a transistor or MOSFET for relay coil driving, a relay with a 5 volt coil, a 1N4148 diode for flyback protection in the simulation, a lamp or equivalent load, an alternating current source for the load side, and suitable connecting wires and ground references for the control side.

The component list is not only a list of parts. It defines the functional chain of the project. The LDR detects the environmental condition. The divider converts that condition into a voltage. The comparator decides whether the condition has crossed the threshold. The driver provides the coil current that the comparator cannot safely provide directly. The relay transfers control action to the load side. The flyback diode protects the switching device from inductive voltage stress.

1

Description

1.1 Introduction

This chapter describes the purpose, background, operating logic, and functional structure of the light activated relay power driver. The project demonstrates how a low voltage sensing circuit can control a higher power electrical load by using a relay as an electromechanical switching device. The central engineering idea is that a weak control signal should not be forced to carry the current of a larger load. Instead, the control signal must command a power switching element, and the power switching element must energize the relay coil.

01 — DESCRIPTION: 5 V “brain” controls isolated 220 V AC “muscle” through a relay

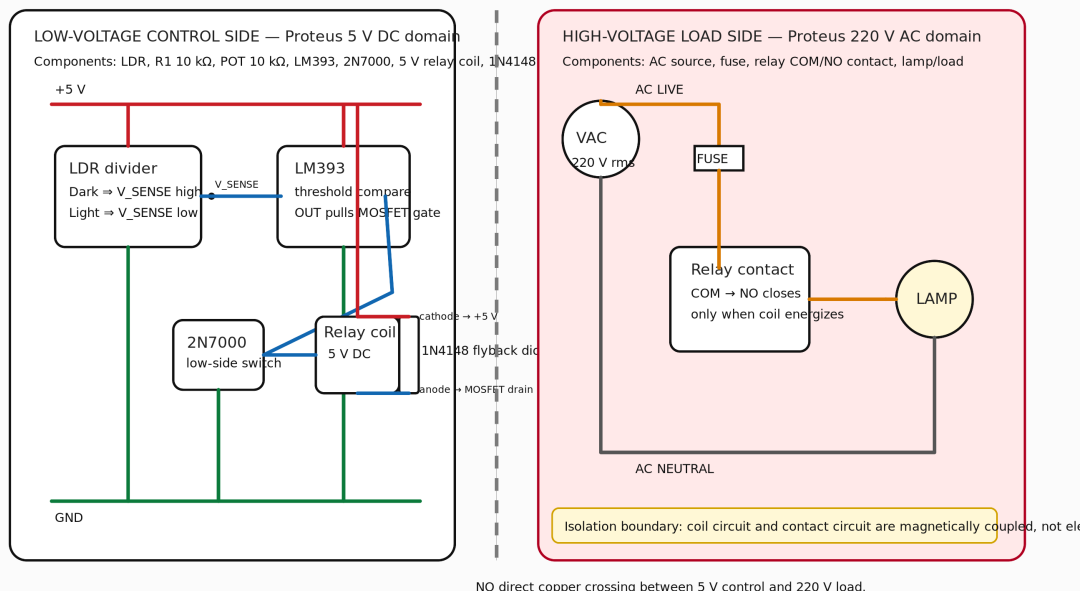


Figure 1.1: Description power driver overview Proteus style.

The complete circuit contains two major domains. The first domain is the low voltage control domain. This domain includes the light dependent resistor, voltage divider,

comparator, pull up resistor, relay driver, relay coil, and flyback diode. The second domain is the load domain. This domain includes the relay contacts and the load being switched. The relay physically separates the coil side from the contact side, allowing a low voltage circuit to control a separate load circuit.

The project is important because relay based switching is common in practical electronics. Sensors, microcontrollers, comparators, and other low power circuits often cannot directly operate lamps, motors, heaters, fans, pumps, or other higher current devices. A relay driver solves this problem when it is designed correctly. Correct design requires not only the relay itself, but also the driver stage and the coil protection stage.

1.2 Project Background

Electronic systems frequently need to respond automatically to changes in the environment. One common example is automatic lighting control. A light dependent resistor can detect whether the surrounding environment is bright or dark. Its resistance changes with light intensity, and this change can be converted into a voltage by placing the LDR in a voltage divider circuit.

However, the output from a light sensing divider is not directly suitable for controlling a high power load. The divider produces a small signal voltage, not a load driving current. For this reason, a control decision stage is required. A comparator is commonly used for this purpose because it compares the sensor voltage with a reference voltage and produces a switching output when the sensor voltage crosses the threshold.

The comparator output still should not be treated as the direct power source for the relay coil. A relay coil usually requires more current than a signal output can safely provide. Therefore, the comparator output must control a driver stage, such as a transistor or a MOSFET. The driver stage supplies the required relay coil current from the 5 volt supply.

The relay coil is an inductor. When current flows through it, magnetic energy is stored in the coil. When the driver suddenly turns off, the magnetic field collapses and the coil attempts to maintain current flow. This produces a reverse voltage spike. If this spike is not controlled, it can damage the driver component or cause simulation instability. A flyback diode is therefore placed across the relay coil to provide a safe current path during turn off.

1.3 Problem Statement

The problem addressed by this project is the safe and reliable control of a higher power load using a low power light sensing circuit. A light dependent resistor circuit can detect brightness or darkness, but it cannot directly switch a lamp or motor. A comparator can make a threshold decision, but it cannot safely supply the current required by a relay coil. A relay can switch a separate load circuit, but its coil must be driven correctly and protected from inductive kickback.

If the relay coil is connected directly to a weak control output, the control device may be overloaded. If the comparator output is left without a proper pull up resistor in an open collector configuration, the output may become undefined. If the flyback diode is omitted, the coil can produce a large reverse voltage when switched off. If the relay contacts are wired incorrectly, the load may remain permanently off, permanently on, or connected in an unsafe way. If the control side and the load side are not clearly separated, the purpose of relay isolation is compromised.

Therefore, the technical problem is not only to make the lamp turn on in darkness. The complete problem is to design a correct sensing, decision, driving, switching, and protection chain that operates reliably in simulation and represents sound circuit design practice.

1.4 Aim of the Project

The aim of this project is to design and simulate a light activated relay power driver in Proteus, where a low voltage LDR based sensing circuit controls a relay, and the relay contacts switch a higher power load when darkness is detected.

The project aims to demonstrate the relationship between sensing, threshold comparison, power driving, relay switching, and inductive protection. It also aims to show that relay operation requires correct connection of the coil side, correct connection of the contact side, correct placement of the flyback diode, and correct interpretation of the switching condition.

1.5 Project Objectives

The first objective is to construct an LDR based voltage divider that produces a measurable voltage change when the light level changes.

The second objective is to provide a reference voltage that defines the switching threshold between the bright condition and the dark condition.

The third objective is to use a comparator to compare the sensor voltage with the reference voltage and generate a control signal.

The fourth objective is to use a driver transistor or MOSFET so that the comparator output does not directly supply the relay coil current.

The fifth objective is to energize a 5 volt relay coil when the dark condition is detected.

The sixth objective is to use the relay contacts to switch a separate load circuit.

The seventh objective is to place a flyback diode across the relay coil with correct polarity so that the inductive voltage spike is suppressed during turn off.

The eighth objective is to verify the expected operation in Proteus by observing the circuit in both bright and dark conditions.

The ninth objective is to identify common wiring and simulation faults, including missing pull up resistance, reversed diode polarity, wrong relay contact use, wrong driver connection, and unstable comparator behavior.

The tenth objective is to produce a formal academic explanation of the circuit that is clear, complete, and technically defensible.

1.6 Working Principle of the System

The system operates by converting a change in light intensity into a switching action. The LDR changes resistance according to the amount of light falling on it. This resistance change alters the voltage at the sensor node. The sensor voltage is compared with a reference voltage. When the sensor voltage crosses the reference voltage in the required direction, the comparator output changes state.

In the recommended dark activated arrangement, the LDR and the fixed resistor are connected so that the sensor voltage rises when the environment becomes dark. If the fixed resistor is connected from the 5 volt supply to the sensor node and the LDR is connected from the sensor node to ground, then the sensor voltage is given by

$$V_{SENSE} = V_{CC} \times \frac{R_{LDR}}{R_{FIXED} + R_{LDR}}$$

When the LDR resistance increases in darkness, the numerator of the fraction increases, and the sensor voltage becomes higher. The comparator can then be connected so that the relay activates when

$$V_{SENSE} > V_{REF}$$

When this condition is satisfied, the comparator output drives the relay driver. The relay driver conducts current through the relay coil. The energized coil creates a magnetic field, which moves the relay contact. The normally open contact closes, and the load circuit is completed. The lamp or load then turns on.

When light returns, the LDR resistance decreases, the sensor voltage falls below the reference voltage, the comparator output changes back, the driver turns off, the relay coil becomes de energized, and the normally open contact opens again. The load circuit is interrupted, and the lamp turns off.

1.7 Low Power Control and High Power Load Concept

The project is based on the distinction between a control signal and a load current. The control signal carries information. The load current carries power. These two roles must not be confused.

The LDR circuit and comparator operate at low voltage and low current. Their purpose is to detect the light condition and produce a decision. They are not designed to carry the current required by a lamp, motor, heater, or other high power device. The load circuit may require a much larger current than the sensing circuit can provide. For this reason, an intermediate switching mechanism is required.

The relay provides this intermediate switching function. The relay coil belongs to the control side, while the relay contacts belong to the load side. The control side

energizes the coil. The load side is switched by the contacts. This arrangement allows the low power circuit to command the high power circuit without directly conducting the high power load current through the sensor or comparator.

This separation can be expressed as a functional chain:

Light condition → Sensor voltage
→ Comparator decision
→ Driver current
→ Relay movement
→ Load switching

Each stage has a different role. The sensor detects. The comparator decides. The driver supplies current. The relay transfers the switching action. The load receives power only when the relay contact path is closed.

1.8 Electromagnetic Switching Concept

A relay operates through electromagnetic switching. The relay coil behaves as an electromagnet when current flows through it. The magnetic field produced by the coil attracts the internal moving contact mechanism. This movement changes the connection between the relay terminals.

In a single pole double throw relay, there are usually three contact terminals: common, normally open, and normally closed. When the coil is not energized, the common terminal is connected to the normally closed terminal. When the coil is energized, the common terminal moves away from the normally closed terminal and connects to the normally open terminal.

For a lamp that must turn on only when darkness is detected, the load should be connected through the common and normally open contacts. In the inactive state, the normally open path is broken and the lamp is off. In the active state, the relay coil is energized, the normally open contact closes, and the lamp turns on.

The electromagnetic switching principle is useful because the coil circuit and the contact circuit do not need to be the same electrical circuit. The coil can be controlled by 5 volts direct current, while the contacts can switch a separate supply appropriate to the load, provided that the relay contact rating is suitable.

1.9 Relay as an Isolation Device

The relay acts as an isolation device because the coil side and the contact side are not normally connected by a direct conductor. The control circuit energizes the coil. The magnetic field moves the contact. The load circuit is switched mechanically through the contact movement.

This isolation is important because the sensing and control components are low voltage devices. They should not be directly connected to a high voltage load circuit. The relay allows the control circuit to influence the load circuit without requiring the sensor

or comparator to carry the load current.

However, relay isolation must not be misunderstood as automatic safety. Correct isolation depends on using a relay with a suitable voltage rating, current rating, insulation rating, and contact spacing. It also depends on correct wiring. The control ground must not be accidentally connected to the high voltage load side. The coil pins must not be confused with the contact pins. The load must be placed in the correct contact path.

In the intended circuit, the 5 volt domain includes the LDR, comparator, driver, relay coil, and flyback diode. The load domain includes the alternating current source, fuse, relay common contact, relay normally open contact, and lamp. The two domains are functionally coupled by relay action, not by direct electrical connection.

1.10 Function of the LDR in the Circuit

The LDR is the light sensing element of the system. Its resistance varies with incident light intensity. In bright conditions, the resistance of the LDR is relatively low. In dark conditions, the resistance of the LDR becomes relatively high. This physical property allows the circuit to transform light level into an electrical variable.

The LDR cannot produce a useful switching decision by itself. It must be connected as part of a voltage divider. In the recommended arrangement,

$$+5V \rightarrow R_{FIXED} \rightarrow V_{SENSE} \rightarrow R_{LDR} \rightarrow GND$$

The voltage at the sensor node is

$$V_{SENSE} = V_{CC} \times \frac{R_{LDR}}{R_{FIXED} + R_{LDR}}$$

This equation shows that the sensor voltage depends on the ratio between the LDR resistance and the total divider resistance. If the LDR resistance increases, the value of the fraction increases. Therefore, in this arrangement, darkness causes the sensor voltage to rise.

The LDR stage is therefore responsible for converting a physical condition into an electrical signal. Its output is not the final control action. It is the input information used by the comparator.

1.11 Function of the Comparator and Control Stage

The comparator stage determines whether the sensed light condition has crossed the selected threshold. It receives two input voltages: the sensor voltage and the reference voltage. The sensor voltage comes from the LDR divider. The reference voltage comes from a potentiometer or another voltage divider.

The switching condition may be written as

$$V_{SENSE} > V_{REF}$$

for dark activated operation in the recommended divider orientation. When this condition is true, the comparator output changes to the state that activates the relay driver. When this condition is false, the comparator output remains in the state that keeps the relay driver off.

The reference voltage is important because it controls the sensitivity of the circuit. A low reference voltage causes the relay to activate at a lower sensor voltage. A higher reference voltage causes the relay to activate only when the sensor voltage becomes higher. If a potentiometer is used, the switching point can be adjusted during simulation.

If the comparator used is an LM393 type, the output requires a pull up resistor. This is because the comparator output can pull the output node low, but it does not actively drive the output high in the same way as a push pull output. The pull up resistor provides a defined high state when the comparator output transistor is not pulling the node low.

The comparator stage is therefore the decision center of the circuit. It does not switch the load directly. It produces the control signal that commands the relay driver stage.

1.12 Function of the Relay Driver Stage

The relay driver stage is required because the comparator output is not intended to supply the relay coil current directly. A relay coil may require tens of milliamperes or more, depending on the relay type. This current demand is greater than what should be taken directly from a sensitive signal output.

The driver stage may use a transistor or a MOSFET. In a low side switching arrangement, the relay coil is connected between the 5 volt supply and the drain or collector of the driver device. The source or emitter is connected to ground. When the driver receives an active signal from the comparator, it conducts and completes the relay coil current path.

The relay coil current is approximately

$$I_{COIL} = \frac{V_{CC}}{R_{COIL}}$$

If the supply voltage is 5 volts and the relay coil resistance is 70 ohms, then

$$I_{COIL} = \frac{5}{70} = 0.0714A$$

This is approximately

$$I_{COIL} = 71.4mA$$

This calculation shows why a driver stage is necessary. The comparator provides the command signal, while the driver device supplies the coil current path.

A gate resistor or base resistor may be included between the comparator output and the driver input. A pull down resistor may also be connected from the driver input to ground. The pull down resistor ensures that the driver remains off when the control

signal is undefined or disconnected.

1.13 Function of the Flyback Diode

The flyback diode protects the driver stage from the voltage produced by the relay coil when the coil current is interrupted. Since the relay coil is an inductor, it resists sudden changes in current. The voltage across an inductor is described by

$$V_L = L \frac{dI}{dt}$$

This equation means that when the current changes very quickly, the induced voltage can become large. When the driver turns off, the coil current attempts to continue flowing. Without a safe path, the coil produces a voltage spike that can damage the transistor or MOSFET and can also cause simulation instability.

The flyback diode is connected across the relay coil with reverse polarity during normal operation. The correct connection is

diode cathode $\rightarrow$ +5V

diode anode $\rightarrow$ coil low side

During normal relay energizing, the diode is reverse biased and does not conduct. During relay turn off, the coil reverses its voltage polarity and forward biases the diode. The coil current then circulates through the diode and decays safely.

The flyback diode does not activate the relay. It does not control the threshold. It does not switch the load. Its role is protection. It gives the stored magnetic energy in the relay coil a controlled discharge path.

1.14 Expected System Behavior

The expected behavior of the system depends on the light level.

In bright conditions, the LDR resistance is low. With the recommended divider orientation, the sensor voltage is below the reference voltage. The comparator output remains in the inactive state. The relay driver remains off. The relay coil is not energized. The normally open contact remains open. The lamp remains off.

In dark conditions, the LDR resistance increases. The sensor voltage rises. When the sensor voltage becomes greater than the reference voltage, the comparator output changes to the active state. The driver device conducts. Current flows through the relay coil. The relay contact moves from its inactive position to its active position. The normally open contact closes. The lamp turns on.

The expected operating sequence is

$$\begin{aligned}
R_{LDR} \uparrow &\Rightarrow V_{SENSE} \uparrow \\
&\Rightarrow V_{SENSE} > V_{REF} \\
&\Rightarrow \text{driver on} \\
&\Rightarrow I_{COIL} > 0 \\
&\Rightarrow \text{relay energized} \\
&\Rightarrow \text{NO contact closed} \\
&\Rightarrow \text{load on}
\end{aligned}$$

When light returns, the reverse sequence occurs:

$$\begin{aligned}
R_{LDR} \downarrow &\Rightarrow V_{SENSE} \downarrow \\
&\Rightarrow V_{SENSE} < V_{REF} \\
&\Rightarrow \text{driver off} \\
&\Rightarrow I_{COIL} = 0 \\
&\Rightarrow \text{relay de energized} \\
&\Rightarrow \text{NO contact open} \\
&\Rightarrow \text{load off}
\end{aligned}$$

The correct system behavior is therefore not only that the lamp turns on. The complete correct behavior is that the lamp turns on only when the sensor voltage crosses the reference threshold, the relay driver energizes the coil, the contact closes, and the load path becomes complete.

1.15 Chapter Summary

This chapter described the complete operating concept of the light activated relay power driver. The system uses an LDR to detect light level, a voltage divider to convert resistance variation into a sensor voltage, a reference voltage to define the switching point, a comparator to decide whether the sensed condition is bright or dark, a driver stage to provide relay coil current, a relay to switch the load circuit, and a flyback diode to protect the driver from inductive voltage stress.

The chapter also established the difference between the control domain and the load domain. The control domain operates at low voltage and performs sensing, decision making, and relay coil driving. The load domain is switched through the relay contacts and may carry higher power. The relay provides functional separation between these domains, but this separation remains valid only when the coil side and contact side are wired correctly.

The next chapter should present the numerical calculations required to justify the component values, including the LDR divider voltage, reference voltage, comparator switching condition, relay coil current, driver requirement, flyback diode suitability, and relay contact current requirement.

Calculation

2.1 Introduction to Circuit Calculations

This chapter presents the electrical calculations required to justify the design of the light-activated relay power driver. The purpose of these calculations is to prove that the selected components can operate within safe and functional electrical limits. The circuit contains two main electrical domains. The first domain is the low-voltage control circuit, which operates from a 5 V direct current supply. This section includes the LDR sensing network, the reference voltage network, the comparator, the relay driver, and the relay coil. The second domain is the load circuit, which is switched by the relay contacts and may represent a higher-power alternating current lamp or motor load.

02 — CALCULATION: exact driver sizing for the Proteus relay + flyback diode circuit

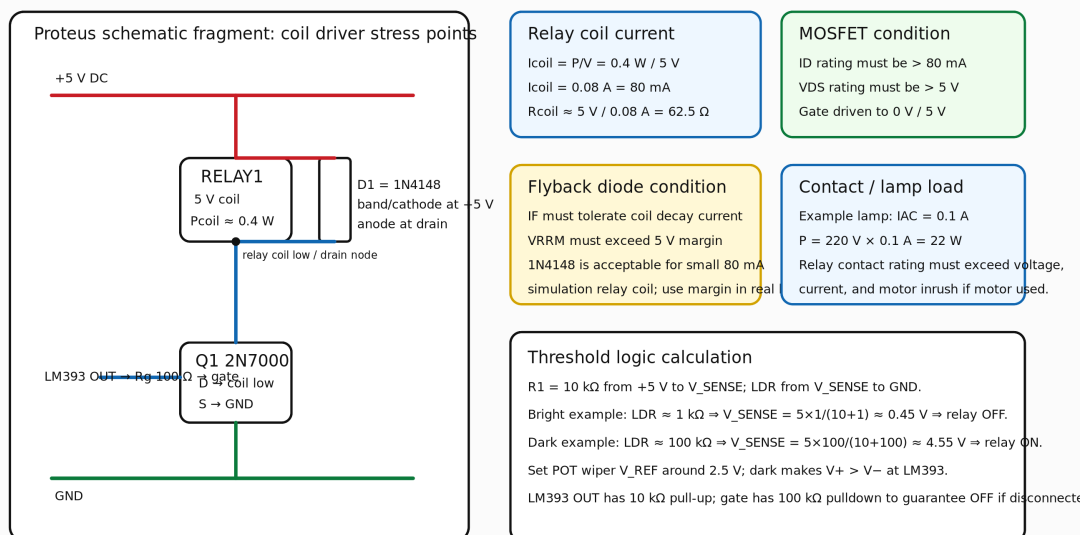


Figure 2.1: *Calculation relay driver values Proteus style.*

The calculation process begins with the selection of the control supply voltage be-

cause every later value depends on this supply. After that, the LDR voltage divider is calculated in both light and dark conditions. These two conditions determine whether the comparator output should remain inactive or activate the relay driver. The comparator threshold is then calculated using an adjustable reference voltage. The relay coil current, relay coil power, driver current requirement, flyback diode requirement, load current, and relay contact rating are then evaluated.

The calculations in this chapter are based on the following design assumptions:

Table 2.1: Design assumptions for the calculation chapter.

Parameter	Assumed Value
Control supply voltage	5 V direct current
Fixed resistor in LDR divider	10 k Ω
Reference potentiometer	10 k Ω
Pull-up resistor	10 k Ω
Relay coil voltage	5 V direct current
Assumed relay coil resistance	70 Ω
Gate resistor	100 Ω
Gate pull-down resistor	100 k Ω
Flyback diode	1N4148
Example load voltage	220 V alternating current
Example lamp power	60 W

These values are suitable for a Proteus-based educational simulation. If the relay model used in Proteus has a different coil resistance, the same formulas must be applied again using the actual relay coil resistance.

2.2 Supply Voltage Selection

The control circuit uses a 5 V direct current supply. This value is selected because many educational relay circuits, comparator circuits, and low-power sensor circuits are designed to operate at 5 V. The relay coil is also selected as a 5 V coil so that the same supply can power the sensing stage, comparator stage, and relay coil stage.

The selected supply voltage is:

$$V_{CC} = 5V$$

The relay coil voltage is also:

$$V_{coil} = 5V$$

The 5 V supply must not be confused with the load-side voltage. The relay coil belongs to the low-voltage control domain, while the relay contacts switch the load domain. The two domains must remain electrically separated. The relay allows the low-voltage control circuit to command the high-power load circuit without directly carrying the load current.

The selected supply voltage must satisfy three conditions. First, it must be sufficient for the comparator circuit. Second, it must match the relay coil voltage. Third, it must be low enough for safe educational simulation and control-side operation. Therefore, 5 V is the appropriate control voltage for this project.

2.3 LDR Voltage Divider Calculation

The LDR is used as a light-sensitive resistor. Its resistance changes according to the amount of light falling on it. In this circuit, the LDR is connected with a fixed resistor to create a voltage divider. The output of this divider is called V_{SENSE} . This voltage is sent to the comparator input.

The selected divider structure is:

$$+5V \rightarrow R_1 \rightarrow V_{SENSE} \rightarrow R_{LDR} \rightarrow GND$$

The voltage divider equation is:

$$V_{SENSE} = V_{CC} \times \frac{R_{LDR}}{R_1 + R_{LDR}}$$

Where:

$$V_{SENSE}$$

is the sensor output voltage,

$$V_{CC}$$

is the 5 V control supply,

$$R_1$$

is the fixed resistor, and

$$R_{LDR}$$

is the resistance of the LDR.

For this design:

$$R_1 = 10k\Omega$$

Therefore:

$$V_{SENSE} = 5V \times \frac{R_{LDR}}{10k\Omega + R_{LDR}}$$

This divider orientation is selected because V_{SENSE} increases when the LDR resistance increases. Since the LDR resistance becomes higher in darkness, this arrangement

makes the sensor voltage rise when the environment becomes dark. This behavior is suitable for a dark-activated relay circuit.

2.4 Light-State Voltage Calculation

In bright light, the resistance of the LDR becomes relatively low. For calculation, assume the LDR resistance in bright light is:

$$R_{LDR-light} = 1k\Omega$$

Using the voltage divider equation:

$$V_{SENSE-light} = 5V \times \frac{1k\Omega}{10k\Omega + 1k\Omega}$$

$$V_{SENSE-light} = 5V \times \frac{1k\Omega}{11k\Omega}$$

$$V_{SENSE-light} = 5V \times 0.0909$$

$$V_{SENSE-light} = 0.455V$$

Therefore, in bright light:

$$V_{SENSE-light} \approx 0.46V$$

This voltage is much lower than the middle reference voltage of approximately 2.5 V. Therefore, if the comparator is wired so that the relay activates only when V_{SENSE} is greater than V_{REF} , the relay remains OFF in bright light.

The expected bright-state result is:

$$V_{SENSE} < V_{REF}$$

Therefore:

$$Relay = OFF$$

and:

$$Lamp = OFF$$

This confirms that the bright condition does not activate the relay.

2.5 Dark-State Voltage Calculation

In darkness, the resistance of the LDR becomes relatively high. For calculation, assume the LDR resistance in darkness is:

$$R_{LDR-dark} = 100k\Omega$$

Using the voltage divider equation:

$$V_{SENSE-dark} = 5V \times \frac{100k\Omega}{10k\Omega + 100k\Omega}$$

$$V_{SENSE-dark} = 5V \times \frac{100k\Omega}{110k\Omega}$$

$$V_{SENSE-dark} = 5V \times 0.909$$

$$V_{SENSE-dark} = 4.545V$$

Therefore, in darkness:

$$V_{SENSE-dark} \approx 4.55V$$

This voltage is higher than the reference voltage of approximately 2.5 V. Therefore, the comparator changes state and activates the relay driver.

The expected dark-state result is:

$$V_{SENSE} > V_{REF}$$

Therefore:

$$Relay = ON$$

and:

$$Lamp = ON$$

This proves that the selected voltage divider arrangement supports dark-activated switching.

2.6 Reference Voltage Calculation

The comparator requires a reference voltage so that it can decide when the LDR voltage is high enough to activate the relay. The reference voltage is produced by a potentiometer connected between the 5 V supply and ground. The potentiometer wiper provides the adjustable reference voltage.

The general reference voltage equation is:

$$V_{REF} = V_{CC} \times \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

If the potentiometer is adjusted to the middle position, then:

$$R_{top} = R_{bottom}$$

Therefore:

$$V_{REF} = 5V \times \frac{R_{bottom}}{R_{bottom} + R_{bottom}}$$

$$V_{REF} = 5V \times \frac{1}{2}$$

$$V_{REF} = 2.5V$$

Thus, the initial reference voltage is:

$$V_{REF} = 2.5V$$

The switching boundary occurs when:

$$V_{SENSE} = V_{REF}$$

Using the LDR divider equation:

$$2.5V = 5V \times \frac{R_{LDR}}{10k\Omega + R_{LDR}}$$

Divide both sides by 5 V:

$$0.5 = \frac{R_{LDR}}{10k\Omega + R_{LDR}}$$

Multiply both sides:

$$0.5(10k\Omega + R_{LDR}) = R_{LDR}$$

$$5k\Omega + 0.5R_{LDR} = R_{LDR}$$

$$5k\Omega = 0.5R_{LDR}$$

$$R_{LDR} = 10k\Omega$$

Therefore, when the reference voltage is set to 2.5 V, the comparator switches when

the LDR resistance is approximately:

$$R_{LDR} = 10k\Omega$$

This is a suitable threshold because the circuit can be adjusted to respond at the desired darkness level by changing the potentiometer position.

2.7 Comparator Switching Condition

The comparator compares the sensor voltage with the reference voltage. In this design, the non-inverting input receives the LDR sensor voltage, and the inverting input receives the reference voltage.

The comparator input assignment is:

$$V_+ = V_{SENSE}$$

$$V_- = V_{REF}$$

The relay should turn ON when the environment becomes dark. Since the selected divider makes V_{SENSE} increase in darkness, the relay activation condition is:

$$V_{SENSE} > V_{REF}$$

When this condition is true, the comparator output becomes HIGH through the pull-up resistor, and the MOSFET gate receives a positive voltage. The MOSFET then turns ON and energizes the relay coil.

The relay OFF condition is:

$$V_{SENSE} < V_{REF}$$

The relay ON condition is:

$$V_{SENSE} > V_{REF}$$

For the calculated values:

Bright state:

$$0.46V < 2.5V$$

Therefore:

$$Relay = OFF$$

Dark state:

$$4.55V > 2.5V$$

Therefore:

$$Relay = ON$$

This confirms that the comparator logic is consistent with the intended operation of the system.

2.8 Pull-Up Resistor Calculation

The comparator output requires a pull-up resistor because the selected comparator output does not actively produce a strong HIGH level by itself. The pull-up resistor connects the output node to the 5 V supply and creates a valid HIGH voltage when the comparator output transistor is not pulling the node LOW.

The selected pull-up resistor is:

$$R_{PU} = 10k\Omega$$

The maximum current through the pull-up resistor occurs when the comparator output is LOW. In that case, the resistor is connected from 5 V to the comparator output transistor path to ground.

The pull-up current is:

$$I_{PU} = \frac{V_{CC}}{R_{PU}}$$

$$I_{PU} = \frac{5V}{10k\Omega}$$

$$I_{PU} = 0.0005A$$

$$I_{PU} = 0.5mA$$

The pull-up resistor power dissipation is:

$$P_{PU} = V_{CC} \times I_{PU}$$

$$P_{PU} = 5V \times 0.5mA$$

$$P_{PU} = 2.5mW$$

This value is very small, so a standard low-power resistor is sufficient. The 10 k Ω pull-up resistor is appropriate because it provides a clear logic HIGH level while keep-

ing the comparator output current low.

2.9 Relay Coil Resistance and Current Calculation

The relay coil is the electromagnetic part of the relay. When current flows through the coil, the relay contact changes position. The coil current must be supplied by the driver transistor or MOSFET, not directly by the comparator.

The relay coil current is calculated using Ohm's law:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

For this design:

$$V_{coil} = 5V$$

$$R_{coil} = 70\Omega$$

Therefore:

$$I_{coil} = \frac{5V}{70\Omega}$$

$$I_{coil} = 0.0714A$$

$$I_{coil} = 71.4mA$$

Therefore, the relay coil requires approximately:

$$I_{coil} \approx 71mA$$

This current is too high to be treated as a simple logic signal. The comparator output is responsible only for controlling the driver. The driver device must carry the relay coil current.

2.10 Relay Coil Power Calculation

The relay coil consumes electrical power when energized. The relay coil power is calculated using:

$$P_{coil} = V_{coil} \times I_{coil}$$

Substituting the calculated current:

$$P_{coil} = 5V \times 0.0714A$$

$$P_{coil} = 0.357W$$

Therefore:

$$P_{coil} \approx 357mW$$

The same result can be confirmed using:

$$P_{coil} = \frac{V_{coil}^2}{R_{coil}}$$

$$P_{coil} = \frac{(5V)^2}{70\Omega}$$

$$P_{coil} = \frac{25}{70}$$

$$P_{coil} = 0.357W$$

This calculation proves that the relay coil is a real load and must be driven by a current-capable switching device. The comparator must not be used as the direct current source for the relay coil.

2.11 Driver Transistor or MOSFET Current Requirement

The relay driver must carry the full relay coil current. Since the calculated relay coil current is approximately 71.4 mA, the driver must safely switch at least this amount of current.

The minimum driver current requirement is:

$$I_{driver-min} = I_{coil}$$

$$I_{driver-min} = 71.4mA$$

A design should not operate exactly at the minimum current rating. A safety margin should be applied. If a margin factor of 2 is used:

$$I_{driver-rated} \geq 2 \times I_{coil}$$

$$I_{driver-rated} \geq 2 \times 71.4mA$$

$$I_{driver-rated} \geq 142.8mA$$

Therefore, the driver should be able to switch at least:

$$I_{driver-rated} \geq 143mA$$

A logic-level N-channel MOSFET is suitable for this role in the simulation because the MOSFET gate requires voltage rather than continuous base current. The MOSFET is connected as a low-side switch. The relay coil is connected between 5 V and the MOSFET drain. The MOSFET source is connected to ground. When the gate receives a HIGH signal, the MOSFET conducts and completes the relay coil current path.

The MOSFET drain current condition is:

$$I_D \geq I_{coil}$$

With safety margin:

$$I_D \geq 143mA$$

The MOSFET drain-source voltage rating must also exceed the control supply voltage. Since the flyback diode clamps the inductive kickback, the MOSFET is protected from a high reverse spike. However, the device should still have a voltage rating comfortably above 5 V.

2.12 Gate or Base Resistor Calculation

In the selected MOSFET driver design, the gate resistor limits transient gate current and reduces sharp switching disturbances. The MOSFET gate behaves mainly as a capacitive input, so it does not require continuous direct current in the way a bipolar transistor base does.

The selected gate resistor is:

$$R_G = 100\Omega$$

The theoretical maximum instant gate current at the moment of switching is:

$$I_{G-peak} = \frac{V_{gate}}{R_G}$$

$$I_{G-peak} = \frac{5V}{100\Omega}$$

$$I_{G-peak} = 0.05A$$

$$I_{G-peak} = 50mA$$

This is only a short transient charging current, not a continuous operating current. After the MOSFET gate capacitance is charged, the steady-state gate current is approximately zero in an ideal model.

If a bipolar transistor is used instead of a MOSFET, the base current must be calculated differently. For a relay coil current of 71.4 mA and a forced current gain of 10:

$$I_B = \frac{I_C}{10}$$

$$I_B = \frac{71.4mA}{10}$$

$$I_B = 7.14mA$$

The base resistor would then be:

$$R_B = \frac{V_{drive} - V_{BE}}{I_B}$$

Assuming:

$$V_{drive} = 5V$$

$$V_{BE} = 0.7V$$

$$R_B = \frac{5V - 0.7V}{7.14mA}$$

$$R_B = \frac{4.3V}{0.00714A}$$

$$R_B = 602\Omega$$

A practical value would be near:

$$R_B \approx 560\Omega$$

or:

$$R_B \approx 680\Omega$$

However, this base-drive calculation shows why the MOSFET is preferred in the selected design. A bipolar transistor would require several milliamperes of base current, while the MOSFET only requires gate voltage and transient gate charge.

2.13 Gate or Base Pull-Down Resistor Justification

The MOSFET gate must not be left floating. A floating gate can cause random relay activation because the MOSFET may partially turn ON due to stored charge, noise, or

simulation instability. A pull-down resistor connects the gate to ground and ensures that the MOSFET remains OFF when the comparator output is inactive or disconnected.

The selected pull-down resistor is:

$$R_{PD} = 100k\Omega$$

When the gate is HIGH at 5 V, the current through the pull-down resistor is:

$$I_{PD} = \frac{V_{gate}}{R_{PD}}$$

$$I_{PD} = \frac{5V}{100k\Omega}$$

$$I_{PD} = 0.00005A$$

$$I_{PD} = 50\mu A$$

The pull-down resistor power is:

$$P_{PD} = V_{gate} \times I_{PD}$$

$$P_{PD} = 5V \times 50\mu A$$

$$P_{PD} = 250\mu W$$

$$P_{PD} = 0.25mW$$

This power is negligible. The 100 k Ω resistor is therefore suitable because it keeps the gate defined without wasting meaningful current. The same principle applies if a transistor base is used. The base should not float, because a floating base can also cause unstable switching.

2.14 Flyback Diode Rating Check

The relay coil is an inductive load. When the driver turns OFF, the magnetic field inside the relay coil collapses. This collapsing magnetic field attempts to keep current flowing and can create a high reverse voltage across the driver. The flyback diode provides a safe path for the coil current during turn-off.

The diode is connected across the relay coil with the following polarity:

$$Cathode \rightarrow +5V$$

Anode → MOSFET drain

During normal relay operation, the diode is reverse-biased and does not conduct. During turn-off, the coil voltage reverses and the diode becomes forward-biased. The stored coil energy then decays safely through the diode and relay coil loop.

The initial flyback diode current is approximately equal to the relay coil current:

$$I_{diode-initial} \approx I_{coil}$$

$$I_{diode-initial} \approx 71.4mA$$

Therefore, the diode must be able to tolerate at least the relay coil current during the decay interval. The reverse voltage rating of the diode must also be higher than the control supply voltage:

$$V_{RRM} > V_{CC}$$

$$V_{RRM} > 5V$$

The stored magnetic energy of the relay coil is:

$$E_{coil} = \frac{1}{2}LI_{coil}^2$$

Where:

$$L$$

is the relay coil inductance. If the relay coil inductance is not provided by the Proteus model or datasheet, the exact stored energy cannot be numerically calculated. However, the design rule remains valid: the diode must support the coil current, reverse voltage, and switching energy produced during coil turn-off.

For the simulation, the 1N4148 diode is acceptable for a small 5 V relay coil when the current is approximately 71 mA. For real hardware, the diode selection must be confirmed using the actual relay coil current and the diode datasheet.

2.15 Load Current Calculation

The relay contact side switches the load circuit. The relay contact current depends on the load power and load voltage. For a resistive lamp load, the current can be calculated using:

$$I_{load} = \frac{P_{load}}{V_{load}}$$

Assume the load is a 60 W lamp connected to a 220 V alternating current supply:

$$P_{load} = 60W$$

$$V_{load} = 220V$$

Therefore:

$$I_{load} = \frac{60W}{220V}$$

$$I_{load} = 0.2727A$$

$$I_{load} \approx 0.273A$$

The equivalent hot resistance of the lamp is:

$$R_{load} = \frac{V_{load}^2}{P_{load}}$$

$$R_{load} = \frac{(220V)^2}{60W}$$

$$R_{load} = \frac{48400}{60}$$

$$R_{load} = 806.7\Omega$$

Therefore, the simulated lamp load has an approximate operating current of 0.273 A and an approximate hot resistance of 807 Ω . This load current is carried only by the relay contacts, not by the comparator, MOSFET gate, LDR, or 5 V control circuit.

2.16 Relay Contact Rating Check

The relay contacts must be rated for the load voltage and load current. The relay coil rating and relay contact rating are different specifications. A 5 V relay coil only means that the coil is energized by 5 V. It does not automatically prove that the contacts can safely switch any load. The contact rating must be checked separately.

The minimum contact current rating must satisfy:

$$I_{contact-rated} > I_{load}$$

Using the calculated lamp current:

$$I_{contact-rated} > 0.273A$$

A safety margin should be applied. If a margin factor of 2 is used:

$$I_{\text{contact-rated}} \geq 2 \times I_{\text{load}}$$

$$I_{\text{contact-rated}} \geq 2 \times 0.273A$$

$$I_{\text{contact-rated}} \geq 0.546A$$

Therefore, the relay contact rating should be at least:

$$I_{\text{contact-rated}} \geq 0.55A$$

For a real circuit, the selected relay should have a contact rating comfortably above this value at the required alternating current voltage. A relay contact rating of several amperes at 220 V alternating current would provide a stronger practical margin for a small lamp load. If the load is a motor, the required contact rating must be higher because motors can draw high starting current.

2.17 Power and Safety Margin Analysis

The power and safety margin analysis confirms that each component operates within a reasonable electrical range. The relay coil consumes the largest amount of power in the 5 V control domain. The pull-up resistor, gate resistor, gate pull-down resistor, and LDR divider consume much smaller amounts of power.

The LDR divider current in bright light is:

$$I_{\text{divider-light}} = \frac{V_{CC}}{R_1 + R_{\text{LDR-light}}}$$

$$I_{\text{divider-light}} = \frac{5V}{10k\Omega + 1k\Omega}$$

$$I_{\text{divider-light}} = \frac{5V}{11k\Omega}$$

$$I_{\text{divider-light}} = 0.455mA$$

The power in the fixed resistor during bright light is:

$$P_{R1-light} = I_{\text{divider-light}}^2 \times R_1$$

$$P_{R1-light} = (0.455mA)^2 \times 10k\Omega$$

$$P_{R1-light} = 2.07mW$$

The LDR divider current in darkness is:

$$I_{divider-dark} = \frac{5V}{10k\Omega + 100k\Omega}$$

$$I_{divider-dark} = \frac{5V}{110k\Omega}$$

$$I_{divider-dark} = 45.5\mu A$$

The power in the fixed resistor during darkness is:

$$P_{R1-dark} = I_{divider-dark}^2 \times R_1$$

$$P_{R1-dark} = (45.5\mu A)^2 \times 10k\Omega$$

$$P_{R1-dark} = 20.7\mu W$$

The relay coil power is:

$$P_{coil} = 357mW$$

The pull-up resistor power is:

$$P_{PU} = 2.5mW$$

The gate pull-down resistor power is:

$$P_{PD} = 0.25mW$$

This analysis shows that the resistor network operates at very low power. The relay coil is the dominant control-side load. Therefore, the relay coil must be switched by a proper driver device.

The safety margin analysis also confirms the following requirements:

Table 2.3: Safety margin requirements.

Item	Required Condition	Design Result
Control supply	Must match relay coil voltage	5 V selected
Comparator threshold	Must separate light and dark states	2.5 V selected
Relay driver current	Must exceed 71.4 mA	Driver must be rated above 143 mA with margin
Flyback diode	Must carry coil decay current	Must support approximately 71.4 mA initial decay current

Item	Required Condition	Design Result
Relay contact current	Must exceed 0.273 A load current	Minimum with margin is 0.55 A
Control and load domains	Must remain separated	Relay contacts provide switching boundary

The design is acceptable when all calculated values remain below the rated limits of the selected components.

2.18 Final Calculation Table

Table 2.5: Final calculation table.

Calculation Item	Formula	Calculated Value	Design Meaning
Control supply voltage	Selected value	5 V	Powers the control circuit and relay coil
Bright LDR voltage	$5V \times \frac{1k\Omega}{10k\Omega+1k\Omega}$	0.455 V	Relay remains OFF
Dark LDR voltage	$5V \times \frac{100k\Omega}{10k\Omega+100k\Omega}$	4.545 V	Relay turns ON
Reference voltage	$5V \times \frac{1}{2}$	2.5 V	Comparator threshold
Switching LDR resistance	Solved from $V_{SENSE} = V_{REF}$	10 k Ω	Boundary between OFF and ON
Pull-up current	$\frac{5V}{10k\Omega}$	0.5 mA	Safe comparator output current
Pull-up power	$5V \times 0.5mA$	2.5 mW	Low resistor power
Relay coil current	$\frac{5V}{70\Omega}$	71.4 mA	Required driver current
Relay coil power	$5V \times 71.4mA$	357 mW	Main control-side load
Minimum driver current with margin	$2 \times 71.4mA$	142.8 mA	Driver should exceed this
Gate peak transient current	$\frac{5V}{100\Omega}$	50 mA	Short gate charging current
Gate pull-down current	$\frac{5V}{100k\Omega}$	50 μ A	Keeps MOSFET OFF
Flyback initial current	Approximately I_{coil}	71.4 mA	Diode must tolerate coil decay current
Lamp current	$\frac{60W}{220V}$	0.273 A	Relay contact load current
Lamp hot resistance	$\frac{(220V)^2}{60W}$	806.7 Ω	Approximate resistive load
Minimum contact rating with margin	$2 \times 0.273A$	0.546 A	Relay contacts must exceed this

2.19 Chapter Summary

This chapter calculated the main electrical values required for the relay-based power driver. The 5 V supply was selected because it matches the relay coil and the low-voltage control circuit. The LDR divider was shown to produce approximately 0.46 V in bright light and approximately 4.55 V in darkness, using the assumed LDR resistance values. With the comparator reference set to 2.5 V, the relay remains OFF in bright light and turns ON in darkness.

The relay coil current was calculated as approximately 71.4 mA, and the relay coil power was calculated as approximately 357 mW. These values prove that the relay coil must be driven by a transistor or MOSFET, not directly by the comparator. The pull-up resistor was calculated to draw only 0.5 mA, which is suitable for producing a valid comparator output level. The MOSFET gate resistor and gate pull-down resistor were justified as stability components for controlled switching.

The flyback diode was shown to be necessary because the relay coil stores magnetic energy and produces a reverse voltage during turn-off. The diode must be connected in reverse across the coil during normal operation so that it conducts only during coil turn-off. The load current was calculated using a 60 W lamp at 220 V, giving approximately 0.273 A. With a safety margin of 2, the relay contact rating should be at least 0.55 A, although a higher rated relay is preferred in practical circuits.

Overall, the calculations confirm that the circuit is electrically consistent. The sensor stage produces a meaningful voltage difference between light and darkness. The comparator threshold separates those conditions. The relay driver supplies the required coil current. The flyback diode protects the driver stage. The relay contacts switch the load current separately from the control circuit. Therefore, the design is mathematically justified before proceeding to the schematic and simulation chapters.

3

Scheme

3.1 Introduction to the Circuit Scheme

This chapter presents the complete circuit scheme of the light activated relay power driver. The purpose of this chapter is to define the physical and electrical arrangement of all components used in the system and to explain how each part is connected inside the Proteus simulation environment.

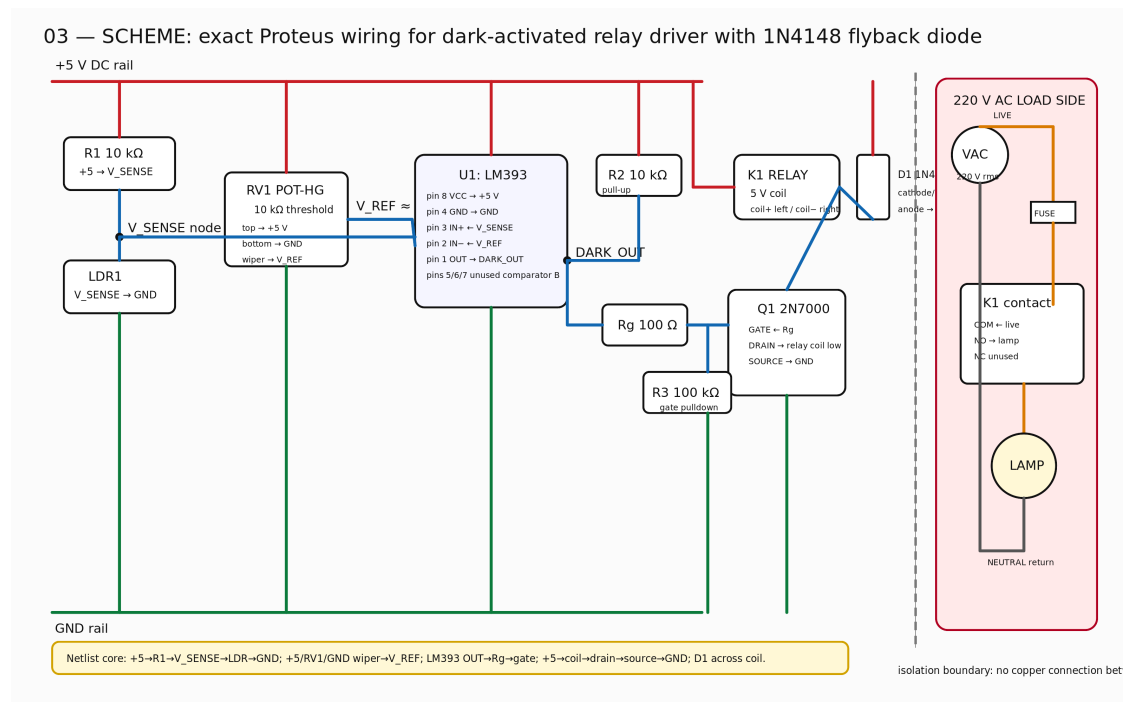


Figure 3.1: Exact Proteus wiring scheme for dark activated operation.

The scheme represents the transition from theoretical design to practical circuit construction. In the previous chapters, the system function and the required calculations were established. In this chapter, those design decisions are converted into a complete wiring structure. The scheme must therefore show not only which components are used, but also why each connection exists and how each electrical node contributes to

the correct operation of the system.

The circuit is divided into two main electrical domains. The first domain is the low voltage control side, which operates from a 5 volt direct current supply. This side contains the light dependent resistor, voltage divider, reference voltage circuit, comparator, driver device, relay coil, and flyback diode. The second domain is the load side, which contains the relay contacts, fuse, alternating current source, and lamp load. These two domains must remain electrically separated except through the electromagnetic action of the relay.

The scheme is designed so that the low power sensing circuit does not directly carry the current of the high power load. Instead, the sensing circuit controls the relay coil through a driver stage, and the relay contacts switch the load circuit. This arrangement allows the circuit to perform controlled switching while preserving the functional separation between the sensing electronics and the load circuit.

The main operating sequence of the scheme is as follows. When the surrounding light level decreases, the resistance of the light dependent resistor changes. This changes the voltage at the sensing node. The comparator compares the sensing voltage with a reference voltage. When the sensing voltage exceeds the reference level, the comparator output activates the relay driver. The relay driver then energizes the relay coil. The energized coil closes the normally open relay contact, and the alternating current load turns on. When the light level increases again, the sensing voltage returns below the reference level, the driver turns off, the relay coil is de energized, and the load turns off.

The correctness of the scheme depends on three essential requirements. First, every component must be placed in the correct electrical domain. Second, every signal path must have a defined source, destination, and reference. Third, the relay coil must be protected against inductive voltage spikes by a correctly connected flyback diode.

3.2 Full Circuit Block Structure

The complete circuit is organized into seven functional blocks. These blocks define the internal structure of the scheme and clarify how the signal moves from light detection to load activation.

The first block is the light sensing block. It contains the light dependent resistor and a fixed resistor. Together, these two components form a voltage divider. The output of this divider is the sensing voltage. This voltage changes according to the amount of light received by the light dependent resistor.

The second block is the reference voltage block. This block contains a potentiometer connected between the 5 volt supply and ground. The wiper terminal of the potentiometer provides an adjustable reference voltage. This voltage determines the threshold at which the system changes from the off state to the on state.

The third block is the comparator block. The comparator receives the sensing voltage at one input and the reference voltage at the other input. It produces an output signal according to the comparison between these two voltages. In this design, the relay must activate when the sensing voltage becomes greater than the reference voltage.

The fourth block is the output conditioning block. Since the comparator output cannot be assumed to produce a valid high level by itself, a pull up resistor is connected from the comparator output node to the 5 volt supply. This resistor ensures that the output node has a defined high state when the comparator output transistor is not pulling the node low.

The fifth block is the relay driver block. This block contains a switching device, preferably an N channel metal oxide semiconductor field effect transistor, used as a low side switch. The driver receives the comparator output signal and controls the relay coil current. This is necessary because the comparator output is a signal output and must not be used as the direct current source for the relay coil.

The sixth block is the relay coil and protection block. The relay coil is connected between the 5 volt supply and the drain of the driver transistor. A flyback diode is connected across the coil. The diode is reverse biased during normal relay operation and conducts only when the driver turns off and the coil produces a reverse voltage spike.

The seventh block is the load switching block. This block contains the relay contacts, fuse, alternating current source, and lamp. The live terminal of the alternating current source passes through the fuse and relay contact before reaching the lamp. The second terminal of the lamp returns to the neutral terminal of the alternating current source. The load side is switched only through the relay contacts and must not be directly connected to the 5 volt control ground.

The block structure can be described in signal order as follows:

Light input changes the resistance of the light dependent resistor. The resistance change produces a change in the sensing voltage. The comparator compares the sensing voltage with the reference voltage. The comparator output controls the driver transistor. The driver transistor controls the relay coil current. The relay coil changes the state of the relay contacts. The relay contacts switch the lamp load.

3.3 Required Proteus Components

The Proteus simulation must include all components needed to represent the sensing, comparison, driving, protection, and load switching functions. The required components are listed below in a formal component structure.

Table 3.1: Required Proteus components.

Component name	Quantity	Electrical domain	Function
Direct current voltage source, 5 volts	1	Control side	Supplies the sensing, comparator, driver, and relay coil circuit
Ground terminal	As required	Control side	Provides the reference potential for the control circuit
Light dependent resistor	1	Control side	Detects the light level

Component name	Quantity	Electrical domain	Function
Fixed resistor, 10 kilo ohms	1	Control side	Forms the sensing voltage divider with the light dependent resistor
Potentiometer, 10 kilo ohms	1	Control side	Produces the adjustable reference voltage
LM393 comparator	1	Control side	Compares the sensing voltage with the reference voltage
Pull up resistor, 10 kilo ohms	1	Control side	Provides a valid high output level for the comparator output
N channel driver transistor	1	Control side	Switches the relay coil current
Gate resistor, 100 ohms	1	Control side	Limits fast current flow into the transistor gate during switching
Gate pull down resistor, 100 kilo ohms	1	Control side	Keeps the transistor off when the control signal is undefined
5 volt relay with single pole double throw contacts	1	Boundary between domains	Allows the control side to switch the load side
1N4148 diode	1	Control side	Protects the driver from relay coil inductive kickback
Alternating current voltage source	1	Load side	Supplies the lamp load circuit
Fuse	1	Load side	Provides load side protection in the simulated scheme
Lamp	1	Load side	Represents the switched load

The components must be selected so that the relay coil voltage matches the control supply. Therefore, a 5 volt relay coil is required when the control supply is 5 volts. A relay with a different coil voltage would not represent the intended operation correctly unless the supply voltage and calculations were also changed.

The relay contact rating must also be suitable for the simulated load voltage and current. In a Proteus educational simulation, the relay may be represented symbolically, but the academic report must still state that real hardware requires a relay with contact ratings greater than the load demand.

3.4 Control Side Circuit Scheme

The control side circuit is the low voltage part of the system. It includes all components that operate from the 5 volt direct current supply. Its purpose is to detect darkness, produce a decision signal, and energize the relay coil through a driver stage.

The control side begins at the 5 volt supply node. This node provides power to the voltage divider, reference voltage circuit, comparator, pull up resistor, and relay coil. The ground node provides the common reference for the sensing circuit, comparator, and driver transistor.

The light sensing branch is connected between the 5 volt supply and ground. The fixed resistor is connected from the 5 volt supply to the sensing node. The light dependent resistor is connected from the sensing node to ground. The sensing node is therefore the midpoint of the divider.

The reference branch is formed by the potentiometer. The upper terminal of the potentiometer is connected to the 5 volt supply. The lower terminal is connected to ground. The wiper terminal becomes the reference voltage node.

The comparator receives the sensing node and reference node as inputs. The non inverting input receives the sensing voltage. The inverting input receives the reference voltage. With this arrangement, the comparator output is intended to activate when the sensing voltage becomes greater than the reference voltage.

The comparator output node is connected to a 10 kilo ohm pull up resistor. The other terminal of this resistor is connected to the 5 volt supply. This creates a defined high state at the comparator output when the comparator output transistor is not pulling the node low.

The comparator output then drives the gate of the transistor through a 100 ohm resistor. The transistor source is connected to ground. The transistor drain is connected to the lower side of the relay coil. The upper side of the relay coil is connected to the 5 volt supply. This makes the transistor a low side switch for the relay coil.

The flyback diode is connected across the relay coil. Its cathode is connected to the 5 volt side of the coil. Its anode is connected to the transistor drain side of the coil. This is the correct polarity because the diode remains non conducting while the coil is normally energized, but conducts when the coil voltage reverses during turn off.

3.5 LDR Sensor Wiring

The light dependent resistor is used as the input sensing element. Its resistance changes according to the amount of light falling on its surface. In this scheme, the light dependent resistor is connected as the lower component in the sensing voltage divider.

The exact wiring is as follows:

The 5 volt supply is connected to one terminal of the fixed 10 kilo ohm resistor. The second terminal of the fixed 10 kilo ohm resistor is connected to the sensing node. The sensing node is connected to one terminal of the light dependent resistor. The second terminal of the light dependent resistor is connected to ground. The sensing node is also connected to the non inverting input of the comparator.

This arrangement produces the sensing voltage according to the voltage divider relation:

$$V_{SENSE} = V_{CC} \times \frac{R_{LDR}}{R_{FIXED} + R_{LDR}}$$

where V_{SENSE} is the sensing voltage, V_{CC} is the 5 volt supply, R_{LDR} is the resistance of the light dependent resistor, and R_{FIXED} is the fixed resistor value.

When the light dependent resistor is exposed to bright light, its resistance decreases. Since it is connected between the sensing node and ground, the sensing voltage decreases. When the light dependent resistor becomes dark, its resistance increases. As a result, the sensing voltage rises toward the supply voltage. This behavior is suitable for a dark activated relay because the comparator can be arranged to switch on when the sensing voltage becomes higher than the reference voltage.

A practical academic report should state that the exact sensing voltage depends on the actual light dependent resistor model used in Proteus. The threshold is therefore adjustable through the potentiometer rather than fixed permanently.

3.6 Reference Voltage Wiring

The reference voltage circuit establishes the threshold against which the sensing voltage is compared. A potentiometer is used so that the switching point can be adjusted during simulation.

The exact wiring is as follows:

The top terminal of the potentiometer is connected to the 5 volt supply. The bottom terminal of the potentiometer is connected to ground. The wiper terminal of the potentiometer is connected to the reference voltage node. The reference voltage node is connected to the inverting input of the comparator.

The reference voltage can be expressed as:

$$V_{REF} = V_{CC} \times k$$

where (k) is the fractional position of the potentiometer wiper. If the wiper is at the midpoint, then:

$$V_{REF} = 5V \times 0.5 = 2.5V$$

The comparator switching condition for this scheme is:

$$V_{SENSE} > V_{REF}$$

When this condition is true, the relay driver should become active. When this condition is false, the relay driver should remain inactive.

The reference voltage must not be left floating. If the potentiometer wiper is not connected correctly, the comparator input may receive an undefined voltage, which can cause unstable switching. Therefore, the reference voltage node must be clearly

labeled and connected directly to the inverting comparator input.

3.7 Comparator Wiring

The comparator is the decision stage of the circuit. It determines whether the system is in the light condition or dark condition by comparing the sensing voltage with the reference voltage.

The comparator wiring is as follows:

The positive supply pin of the comparator is connected to the 5 volt supply. The ground pin of the comparator is connected to the control ground. The non inverting input is connected to the sensing voltage node. The inverting input is connected to the reference voltage node. The output pin is connected to the dark output node. The dark output node is connected to the pull up resistor and then to the transistor gate circuit.

The intended comparator logic is:

$$V_{SENSE} > V_{REF} \Rightarrow \text{Output active}$$

$$V_{SENSE} < V_{REF} \Rightarrow \text{Output inactive}$$

This input arrangement is important. If the sensing voltage and reference voltage are connected to the opposite inputs, the relay behavior will be reversed. In that incorrect arrangement, the lamp may turn on in light and turn off in darkness. Therefore, the comparator input polarity must match the intended behavior of the system.

The comparator should be understood as a decision device, not as a power device. Its output is used only to command the driver transistor. It must not be used to supply relay coil current directly.

3.8 Comparator Output Pull Up Wiring

The pull up resistor is a necessary part of the comparator output stage. In this scheme, a 10 kilo ohm resistor is connected from the comparator output node to the 5 volt supply.

The exact wiring is as follows:

One terminal of the 10 kilo ohm pull up resistor is connected to the 5 volt supply. The second terminal of the pull up resistor is connected to the comparator output node. The same comparator output node is connected to the driver input path.

The function of the pull up resistor is to define the high output state. Without this resistor, the comparator output node may not rise to a valid voltage level. A floating output node can cause the transistor to switch unpredictably, prevent the relay from activating, or produce unstable operation near the threshold.

The pull up resistor value of 10 kilo ohms is suitable for a simulation level signal because it provides a stable high state without drawing excessive current. The pull up current can be estimated by:

$$I_{PULLUP} = \frac{V_{CC}}{R_{PULLUP}}$$

With a 5 volt supply and 10 kilo ohm resistor:

$$I_{PULLUP} = \frac{5V}{10000\Omega} = 0.0005A$$

$$I_{PULLUP} = 0.5mA$$

This current is small enough for the control circuit while still defining the output voltage clearly.

3.9 Relay Driver Wiring

The relay driver is required because the relay coil needs more current than the comparator output should provide. The driver device acts as an electronic switch that controls the relay coil current.

In this scheme, an N channel transistor is used as a low side switch. A suitable small signal N channel device may be used in Proteus, provided that it can handle the relay coil current in the simulation.

The exact wiring is as follows:

The comparator output node is connected to one terminal of the 100 ohm gate resistor. The second terminal of the 100 ohm gate resistor is connected to the transistor gate. The transistor source is connected to ground. The transistor drain is connected to the relay coil low side node. A 100 kilo ohm resistor is connected between the transistor gate and ground. The relay coil high side is connected to the 5 volt supply. The relay coil low side is connected to the transistor drain.

The gate resistor limits sudden current flow into the gate during switching transitions. The gate pull down resistor keeps the transistor off when the comparator output is undefined, disconnected, or temporarily inactive during simulation startup.

The driver operation can be summarized as follows. When the comparator output becomes high, the transistor gate receives a positive voltage. The transistor turns on and creates a current path from the 5 volt supply, through the relay coil, through the transistor, and finally to ground. When the comparator output becomes low, the transistor turns off, the coil current stops, and the relay returns to its inactive contact state.

The relay coil current path during activation is:

5 volt supply to relay coil high side. Relay coil high side through relay coil winding. Relay coil low side to transistor drain. Transistor drain to transistor source. Transistor source to ground.

This current path must be continuous. If any part of this path is missing, the relay will not energize.

3.10 Relay Coil Wiring

The relay coil belongs to the low voltage control side of the system. It must be powered by the same 5 volt supply used by the control circuit, provided that the selected relay coil is rated for 5 volts.

The exact relay coil wiring is as follows:

One terminal of the relay coil is connected to the 5 volt supply. The second terminal of the relay coil is connected to the transistor drain. The transistor source is connected to ground. The flyback diode is connected in parallel with the relay coil.

The relay coil current is determined by the coil voltage and coil resistance:

$$I_{COIL} = \frac{V_{COIL}}{R_{COIL}}$$

If the relay coil is 5 volts and the coil resistance is 70 ohms, then:

$$I_{COIL} = \frac{5V}{70\Omega} = 0.0714A$$

$$I_{COIL} \approx 71.4mA$$

The coil power is:

$$P_{COIL} = V_{COIL} \times I_{COIL}$$

$$P_{COIL} = 5V \times 0.0714A = 0.357W$$

The driver transistor must therefore be able to switch at least the calculated coil current with a suitable safety margin. The comparator output alone should not be used as the coil current path.

The coil pins must not be confused with the contact pins. The coil pins belong to the 5 volt control side. The contact pins belong to the switched load side.

3.11 Flyback Diode Wiring

The flyback diode is connected directly across the relay coil to protect the driver transistor from the voltage spike generated when the relay coil turns off.

The exact wiring is as follows:

The cathode of the 1N4148 diode is connected to the 5 volt side of the relay coil. The anode of the 1N4148 diode is connected to the transistor drain side of the relay coil.

This polarity is essential. During normal relay operation, the 5 volt side of the coil is more positive than the transistor drain side. Therefore, the diode is reverse biased and does not conduct. When the transistor turns off, the magnetic field in the relay coil collapses and the coil attempts to keep current flowing. This causes the voltage across the coil to reverse. The diode then becomes forward biased and provides a safe decay

path for the coil current.

The protection action can be explained by the inductor voltage relation:

$$V_L = L \frac{di}{dt}$$

where V_L is the voltage across the coil, (L) is the coil inductance, and $\frac{di}{dt}$ is the rate of current change. When the transistor turns off suddenly, the current tries to fall rapidly. This creates a large voltage. The flyback diode reduces this voltage by allowing the current to circulate through the coil and diode until the stored magnetic energy is dissipated.

The stored magnetic energy in the relay coil can be represented by:

$$E = \frac{1}{2}LI^2$$

where (E) is the stored energy, (L) is the inductance of the coil, and I is the coil current before turn off. The flyback diode provides a controlled path for this energy. Without the diode, the voltage spike may disturb the simulation, damage the driver device in real hardware, or cause unstable switching behavior.

A reversed diode would be incorrect because it would conduct during normal coil energizing and could create a direct current path from the 5 volt supply through the diode and transistor path. Therefore, the diode orientation must be verified before running the simulation.

3.12 Relay Contact Side Wiring

The relay contact side is the switching section of the load circuit. It is physically controlled by the relay coil but electrically separated from the 5 volt control side.

A single pole double throw relay has three contact terminals:

Common terminal. Normally open terminal. Normally closed terminal.

For this project, the lamp must turn on only when darkness activates the relay. Therefore, the normally open terminal must be used.

The exact contact side wiring is as follows:

The live terminal of the alternating current source is connected to the fuse. The output terminal of the fuse is connected to the relay common terminal. The relay normally open terminal is connected to the first terminal of the lamp. The second terminal of the lamp is connected to the neutral terminal of the alternating current source. The normally closed terminal is left unused.

When the relay is inactive, the common terminal is not connected to the normally open terminal. The load circuit is open, and the lamp remains off. When the relay coil is energized, the common terminal connects to the normally open terminal. The load circuit becomes complete, and the lamp turns on.

The normally closed terminal should not be used for the required dark activated lamp behavior. If the lamp is connected through the normally closed contact, it will be on when the relay is inactive and will turn off when the relay activates. That would be

the opposite of the required behavior.

3.13 AC Load Wiring

The alternating current load wiring represents the high power side of the circuit. Even in simulation, it must be drawn as a separate circuit from the 5 volt control side.

The correct load current path is:

Alternating current live terminal to fuse. Fuse to relay common terminal. Relay common terminal to relay normally open terminal when activated. Relay normally open terminal to lamp terminal one. Lamp terminal two to alternating current neutral terminal.

The load current can be calculated using:

$$I_{LOAD} = \frac{P_{LOAD}}{V_{AC}}$$

If the lamp power is 60 watts and the alternating current supply is 220 volts, then:

$$I_{LOAD} = \frac{60W}{220V} = 0.2727A$$

$$I_{LOAD} \approx 0.273A$$

The relay contact rating must be greater than the load current. A real circuit must also consider lamp inrush current because the initial current of a cold filament lamp can be higher than the steady state current. In the academic report, the relay contact rating should therefore be stated with a margin above the calculated load current.

The alternating current side must never be connected to the 5 volt ground. The relay contacts act as an isolated switch and must not be treated as part of the control ground network.

3.14 Fuse and Load Protection Placement

The fuse is placed in series with the live side of the alternating current load circuit. Its purpose is to interrupt the load circuit if excessive current flows.

The correct placement is:

Alternating current live terminal to fuse. Fuse to relay common terminal. Relay normally open terminal to lamp. Lamp to alternating current neutral terminal.

Placing the fuse on the live conductor before the relay common terminal ensures that the downstream load path is protected. If a fault occurs in the relay contact path or lamp branch, the fuse can open the live supply path.

The fuse rating should be selected above the normal operating current of the load but below the maximum safe current of the wiring and relay contact path. For a purely illustrative simulation with a 60 watt lamp at 220 volts, the steady state current is approximately 0.273 amperes. A practical fuse rating would need to consider inrush be-

havior and safety requirements. The academic report should not claim that one fixed fuse value is universally correct without knowing the real lamp, supply, wiring, and protection standard.

In the Proteus scheme, the fuse also improves the clarity of the load side because it shows that load switching is not only a functional issue but also a protection issue.

3.15 Grounding and Isolation Rules

Grounding and isolation are central to the correctness of the scheme. The 5 volt control ground is used only for the low voltage circuit. It is connected to the light dependent resistor branch, potentiometer branch, comparator ground pin, transistor source, and gate pull down resistor.

The alternating current load circuit must not share this ground. The alternating current live and neutral terminals form their own load circuit. The relay contact behaves as a switch inside that load circuit. The relay coil and relay contact are functionally related but electrically separated.

The isolation rules are as follows.

The 5 volt supply ground must connect only to low voltage control components. The alternating current neutral must return only to the alternating current source. The relay coil pins must remain on the 5 volt side. The relay common, normally open, and normally closed pins must remain on the load side. The flyback diode must be connected only across the relay coil, not across the load contacts. The load fuse must be placed in the alternating current live path, not in the 5 volt control path. No wire should connect alternating current live or neutral to the 5 volt control ground.

These rules prevent false isolation. A relay only provides useful separation if the external wiring preserves that separation.

3.16 Complete Wire Connection Table

The complete scheme can be verified using the following wire connection table.

Table 3.3: Complete wire connection table.

Connection number	From	To	Node name	Purpose
1	5 volt supply positive	Fixed resistor R1 terminal one	5V	Supplies the sensing divider
2	Fixed resistor R1 terminal two	Light dependent resistor terminal one	VSENSE	Forms the sensing node
3	Light dependent resistor terminal two	Control ground	GND	Completes the sensing divider

Connection number	From	To	Node name	Purpose
4	VSENSE node	Comparator non inverting input	VSENSE	Sends sensing voltage to comparator
5	5 volt supply positive	Potentiometer top terminal	5V	Supplies reference divider
6	Potentiometer bottom terminal	Control ground	GND	Completes reference divider
7	Potentiometer wiper	Comparator inverting input	VREF	Sends threshold voltage to comparator
8	Comparator positive supply pin	5 volt supply positive	5V	Powers comparator
9	Comparator ground pin	Control ground	GND	Provides comparator reference
10	5 volt supply positive	Pull up resistor terminal one	5V	Supplies output pull up
11	Pull up resistor terminal two	Comparator output	DARK OUT	Defines comparator high state
12	Comparator output	Gate resistor terminal one	DARK OUT	Sends output signal to driver
13	Gate resistor terminal two	Transistor gate	GATE	Controls transistor
14	Transistor gate	Gate pull down resistor terminal one	GATE	Prevents floating gate
15	Gate pull down resistor terminal two	Control ground	GND	Holds gate low when inactive
16	Transistor source	Control ground	GND	Completes low side switch path
17	5 volt supply positive	Relay coil terminal one	5V	Supplies relay coil
18	Relay coil terminal two	Transistor drain	COIL LOW	Allows transistor to switch coil
19	Diode cathode	Relay coil 5 volt side	5V	Sets flyback diode polarity
20	Diode anode	Relay coil low side	COIL LOW	Provides flyback current path
21	Alternating current live	Fuse terminal one	AC LIVE	Begins load path
22	Fuse terminal two	Relay common terminal	FUSED LIVE	Protects switched live path

Connection number	From	To	Node name	Purpose
23	Relay normally open terminal	Lamp terminal one	SWITCHED LIVE	Supplies lamp when relay activates
24	Lamp terminal two	Alternating current neutral	AC NEUTRAL	Completes load return path
25	Relay normally closed terminal	Not connected	NC	Not used in dark activated operation

This table is the wiring authority for the scheme. If the schematic diagram and this table disagree, the schematic must be corrected until every connection matches the table.

3.17 Expected Node Labels in Proteus

Node labels improve readability and make debugging easier during simulation. The following node labels should be used in the Proteus schematic.

Table 3.5: Expected Proteus node labels.

Node label	Meaning	Expected location
5V	Positive control supply	Direct current supply positive terminal
GND	Control ground	Low voltage reference node
VSENSE	LDR divider output	Between fixed resistor and light dependent resistor
VREF	Reference voltage	Potentiometer wiper
DARK OUT	Comparator output	Comparator output and pull up resistor junction
GATE	Driver control input	After the gate resistor
COIL LOW	Relay coil low side	Relay coil and transistor drain junction
AC LIVE	Alternating current live	Alternating current source live terminal
FUSED LIVE	Protected live line	Between fuse and relay common terminal
SWITCHED LIVE	Relay switched live output	Between normally open contact and lamp
AC NEUTRAL	Alternating current neutral	Lamp return path

The use of these labels makes the scheme academically clear because each important node has a defined electrical identity. It also allows simulation probes to be placed accurately.

The most important probe points are VSENSE, VREF, DARK OUT, GATE, COIL LOW, SWITCHED LIVE, and lamp voltage. These points allow the designer to confirm the complete chain of operation from light detection to load activation.

3.18 Final Full Schematic Description

The final schematic consists of a 5 volt light controlled relay driver and a separately switched load circuit. The low voltage control circuit begins with a light dependent resistor voltage divider. A fixed 10 kilo ohm resistor is connected from the 5 volt supply to the sensing node, and the light dependent resistor is connected from the sensing node to ground. The sensing node increases in voltage when darkness increases the resistance of the light dependent resistor.

A 10 kilo ohm potentiometer is connected between the 5 volt supply and ground to create an adjustable reference voltage. The potentiometer wiper is connected to the reference input of the comparator. The comparator compares the sensing voltage with the reference voltage. The sensing voltage is connected to the non inverting input, and the reference voltage is connected to the inverting input. This arrangement makes the comparator activate when darkness causes the sensing voltage to exceed the reference voltage.

The comparator output is connected to a 10 kilo ohm pull up resistor tied to the 5 volt supply. This creates a valid high output level for the driver stage. The comparator output then passes through a 100 ohm gate resistor to the gate of the N channel driver transistor. A 100 kilo ohm resistor connects the gate to ground so that the transistor remains off when no valid drive signal is present.

The relay coil is connected between the 5 volt supply and the drain of the driver transistor. The transistor source is connected to ground. When the transistor gate receives a high signal, the transistor conducts and allows current to flow through the relay coil. The relay then changes contact state.

A 1N4148 flyback diode is connected across the relay coil. The diode cathode is connected to the 5 volt side of the coil, and the diode anode is connected to the transistor drain side of the coil. During normal operation, the diode is reverse biased. During turn off, the diode conducts the coil decay current and protects the driver transistor from the inductive voltage spike.

The load side is wired through the relay contacts. The alternating current live terminal is connected to a fuse. The fuse output is connected to the relay common terminal. The relay normally open terminal is connected to the first terminal of the lamp. The second terminal of the lamp is connected to the alternating current neutral terminal. When the relay is inactive, the normally open contact is open and the lamp is off. When the relay is energized, the normally open contact closes and the lamp turns on.

The final schematic therefore satisfies the intended system function. It allows darkness to activate a relay controlled load while preserving the separation between the low voltage sensing circuit and the higher voltage load circuit.

3.19 Chapter Summary

This chapter defined the complete circuit scheme for the light activated relay power driver. The system was divided into a low voltage control side and a separate load side. The control side contains the light dependent resistor voltage divider, adjustable ref-

erence voltage, comparator, pull up resistor, relay driver, relay coil, and flyback diode. The load side contains the relay contacts, fuse, alternating current source, and lamp.

The chapter established the exact wiring of each section. The light dependent resistor and fixed resistor form the sensing voltage divider. The potentiometer creates the adjustable reference voltage. The comparator compares the sensing voltage with the reference voltage. The pull up resistor creates a valid comparator output level. The driver transistor switches the relay coil current. The flyback diode protects the driver from inductive kickback. The relay contacts switch the lamp through the normally open contact path.

The chapter also clarified the required node labels, complete wire connection table, grounding rules, and isolation rules. The most important structural requirement is that the 5 volt control side and the alternating current load side must remain electrically separated. The relay coil belongs to the control side, while the relay contacts belong to the load side.

The completed scheme provides a clear and technically defensible basis for the next chapter, where the Proteus simulation will verify the expected behavior in both light and dark conditions.

4

Simulation

4.1 Introduction to Proteus Simulation

Proteus simulation is used in this project to verify the functional behavior of the light activated relay driver before considering any physical implementation. The simulation allows the circuit to be tested under controlled conditions, where the light dependent resistor, comparator, relay coil, flyback diode, and alternating current load can be observed without exposing the circuit to physical wiring errors or unsafe high voltage handling.

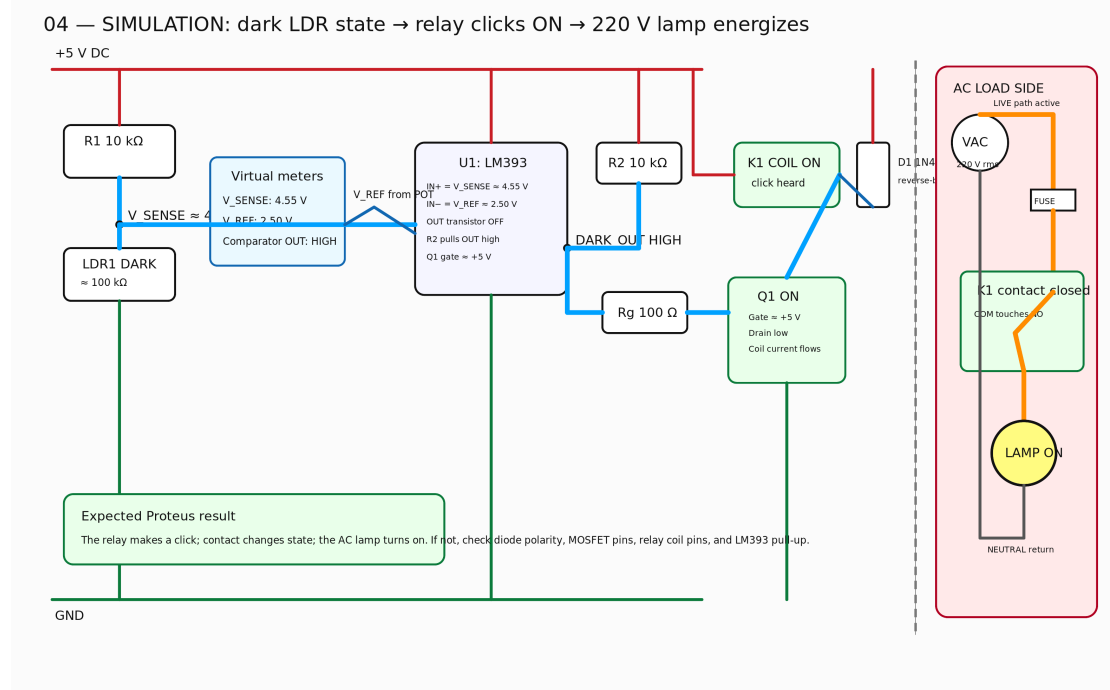


Figure 4.1: Simulation dark state relay on Proteus style.

The purpose of this chapter is to demonstrate that the proposed circuit operates according to its intended electrical logic. When the light level is high, the relay should remain deactivated and the lamp should remain off. When the light level decreases

below the selected threshold, the relay driver should energize the relay coil, the relay contact should close, and the lamp should turn on. The simulation therefore provides the main evidence that the circuit design is functionally correct.

The simulation also verifies that the low voltage control side and the higher voltage load side remain separated by the relay. The control side contains the light sensing circuit, comparator, driver transistor or MOSFET, relay coil, and flyback diode. The load side contains the alternating current source, fuse, relay contact, and lamp. Correct simulation requires that these two domains interact only through the relay mechanism and not through any direct electrical connection between the control ground and the load conductor.

Proteus is especially suitable for this project because it can represent both analogue sensing behavior and electromechanical relay switching behavior. It allows voltage probes, current probes, and waveform graphs to be placed at important nodes so that the operation of the circuit can be examined in a measurable way. This chapter therefore does not only state that the circuit works, but also defines how the circuit must be tested, observed, and accepted.

4.2 Simulation Objectives

The first objective of the simulation is to confirm that the light dependent resistor produces a variable sensing voltage according to the simulated light condition. The sensing voltage must change in a direction that agrees with the selected voltage divider orientation. In the chosen arrangement, the fixed resistor is connected from the 5 volt supply to the sensing node, and the light dependent resistor is connected from the sensing node to ground. Therefore, when the light dependent resistor resistance increases in darkness, the sensing voltage also increases.

The second objective is to verify that the comparator changes its output state when the sensing voltage crosses the reference voltage. The comparator must not behave randomly or remain in an undefined state. Its output must be supported by a pull up resistor so that the high output state becomes electrically valid.

The third objective is to confirm that the comparator output can control the relay driver stage. The relay coil should not be driven directly from the comparator output. Instead, the comparator output should control the gate or base of a driver device, and the driver device should provide the current path for the relay coil.

The fourth objective is to verify that the relay coil energizes only when the simulated darkness condition is reached. The relay must remain off during the bright condition and must activate during the dark condition. This behavior confirms that the sensing, comparison, and driving stages are logically coordinated.

The fifth objective is to verify that the relay contact switches the lamp circuit correctly. The common contact and normally open contact must be used so that the lamp remains off when the relay is not energized and turns on when the relay is energized.

The sixth objective is to verify that the flyback diode is correctly connected across the relay coil. The diode must not conduct during normal coil energization, but it must provide a safe current path when the relay coil is switched off.

The seventh objective is to confirm that all relevant voltages and currents remain within expected values. This includes the sensing voltage, reference voltage, comparator output voltage, relay coil voltage, relay coil current, and lamp voltage.

4.3 Proteus Project Setup

A new Proteus project should be created for the simulation of the relay based power driver circuit. The project should be configured as a schematic simulation project because the main purpose is to test circuit behavior rather than to design a printed circuit board layout. The schematic workspace should be kept organized so that the control side and load side are visually separated.

The left side of the schematic should contain the low voltage control circuit. This area should include the 5 volt direct current source, ground reference, light dependent resistor, fixed resistor, potentiometer, comparator, pull up resistor, driver device, relay coil, and flyback diode. The right side of the schematic should contain the load switching circuit. This area should include the alternating current source, fuse, relay contact terminals, and lamp.

The relay should be placed between the two domains because it is the boundary component that connects the control decision to the load switching action. The coil pins should remain on the low voltage side, while the common, normally open, and normally closed contact pins should remain on the load side. This visual separation helps prevent the most serious structural mistake, which is confusing relay coil wiring with relay contact wiring.

All important circuit nodes should be named before running the simulation. Recommended node names include V_{CC} , (GND), V_{SENSE} , V_{REF} , $DARKOUT$, $COIL_{LOW}$, AC_{LIVE} , $AC_{NEUTRAL}$, (COM), and (NO). Named nodes improve readability and make measurement results easier to interpret.

The simulation time should be long enough to observe both the steady bright condition and the steady dark condition. If the light dependent resistor value is changed manually during simulation, the circuit should be allowed enough time to settle after each change.

4.4 Component Placement Procedure

The first components to be placed are the 5 volt direct current source and the ground reference. These define the supply for the sensing and control side of the circuit. The ground reference must be used consistently throughout the control side.

The next components to be placed are the fixed resistor and the light dependent resistor. These two components form the sensing voltage divider. The fixed resistor should be connected between the 5 volt supply and the sensing node. The light dependent resistor should be connected between the sensing node and ground. The sensing node should be labelled V_{SENSE} .

The potentiometer should then be placed to generate the adjustable reference voltage. The upper terminal of the potentiometer should be connected to the 5 volt supply,

the lower terminal should be connected to ground, and the wiper terminal should be labelled V_{REF} . This reference voltage determines the light threshold at which the relay changes state.

The comparator should be placed after the sensing and reference networks. The non inverting input should receive V_{SENSE} , and the inverting input should receive V_{REF} , provided that the design objective is to activate the relay when the sensing voltage becomes greater than the reference voltage. The comparator supply pin should be connected to the 5 volt supply, and the comparator ground pin should be connected to the control ground.

A pull up resistor must be placed from the comparator output node to the 5 volt supply. This output node should be labelled $DARK_{OUT}$. Without this resistor, the comparator output cannot reliably provide a high state.

The relay driver device should be placed between the comparator output and the relay coil. If a MOSFET is used, the comparator output should be connected to the MOSFET gate through a small gate resistor. The MOSFET source should be connected to ground, and the MOSFET drain should be connected to the low side of the relay coil. A high value pull down resistor should be connected from the gate to ground so that the MOSFET remains off when the control signal is undefined.

The relay coil should be connected between the 5 volt supply and the driver switching node. The flyback diode should be connected directly across the coil. The cathode of the diode should be connected to the 5 volt side of the coil, and the anode should be connected to the driver side of the coil. This polarity keeps the diode reverse biased during normal operation.

The load side should be placed last. The alternating current live conductor should pass through a fuse and then connect to the relay common contact. The normally open contact should connect to one terminal of the lamp. The other terminal of the lamp should return to the alternating current neutral conductor. This arrangement ensures that the lamp turns on only when the relay coil is energized.

4.5 Power Supply Configuration

The control supply should be configured as a 5 volt direct current source. This supply powers the light sensing circuit, comparator circuit, relay coil, and driver stage. The 5 volt value is selected because the relay coil and comparator circuit are designed to operate in the low voltage control domain.

The polarity of the control supply must be verified before the simulation is started. The positive terminal must be connected to the V_{CC} rail, and the negative terminal must be connected to the control ground. If the supply polarity is reversed, the comparator, relay coil, and driver stage will not operate correctly and may show invalid simulation behavior.

The alternating current load supply should be configured according to the intended simulated load condition. If the project represents a mains lamp, the alternating current source may be configured with a nominal value such as 220 volts root mean square for simulation purposes. However, the report must clearly state that this value repre-

sents the load side only and must not be connected to the 5 volt control ground.

The control supply and load supply must remain separated. The relay contact is the switching element between the load supply and the lamp, while the relay coil belongs to the 5 volt side. The simulation is correct only if the relay performs the switching action without any direct conductor joining the high voltage load circuit to the low voltage control circuit.

The mathematical control supply condition is:

$$V_{CC} = 5V$$

The relay coil voltage during activation should approach:

$$V_{coil} \approx V_{CC}$$

The relay coil current is determined by:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

This current must be supplied through the driver device, not directly through the comparator output.

4.6 LDR Light Level Configuration

The light dependent resistor must be configured so that its resistance changes according to the simulated light condition. In bright light, the resistance of the light dependent resistor is relatively low. In darkness, the resistance of the light dependent resistor is relatively high. The exact values depend on the selected component model, but the functional behavior must remain consistent.

In the selected voltage divider arrangement, the sensing voltage is calculated by:

$$V_{SENSE} = V_{CC} \times \frac{R_{LDR}}{R_{fixed} + R_{LDR}}$$

Because R_{LDR} is in the lower position of the divider, an increase in R_{LDR} causes an increase in V_{SENSE} . Therefore, darkness should raise the sensing voltage, and brightness should lower the sensing voltage.

During bright simulation, the expected condition is:

$$R_{LDR} = low$$

$$V_{SENSE} < V_{REF}$$

During dark simulation, the expected condition is:

$$R_{LDR} = high$$

$$V_{SENSE} > V_{REF}$$

The light level should be adjusted gradually in Proteus so that the switching threshold can be observed. A sudden unrealistic change may still switch the relay, but a gradual change provides better evidence of comparator threshold behavior.

The report should record at least two light conditions. The first should represent bright light, where the relay remains off. The second should represent darkness, where the relay turns on. If available, an intermediate condition near the threshold may also be recorded to show the exact switching point.

4.7 Comparator Threshold Configuration

The comparator threshold is established by the reference voltage at the potentiometer wiper. The reference voltage determines the boundary between the bright state and the dark state. The comparator compares V_{SENSE} against V_{REF} .

For the selected input arrangement, the relay should activate when:

$$V_{SENSE} > V_{REF}$$

The relay should deactivate when:

$$V_{SENSE} < V_{REF}$$

If the potentiometer is set near the middle of the 5 volt supply, the reference voltage is approximately:

$$V_{REF} \approx 2.5V$$

This value is not mandatory, but it is a practical starting point for simulation. The final threshold may be adjusted according to the simulated light dependent resistor values.

The comparator output should be connected to a pull up resistor. The pull up resistor provides a valid high voltage level when the comparator output transistor is not pulling the output low. A common value for the pull up resistor is:

$$R_{pullup} = 10k\Omega$$

The comparator output behavior can be interpreted as follows. If V_{SENSE} is below V_{REF} , the output should remain in the state that keeps the relay driver off. If V_{SENSE} rises above V_{REF} , the output should change to the state that turns the relay driver on. If the output behavior is reversed, the comparator input terminals should be checked and corrected.

4.8 Relay Coil Simulation Setup

The relay coil should be configured as a 5 volt coil because the control circuit uses a 5 volt supply. The coil must be connected between the positive 5 volt rail and the driver switching node. The driver device completes the coil current path to ground when it turns on.

The expected coil current is calculated by:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

If the relay coil resistance is 70Ω , and the coil voltage is approximately (5V), the expected coil current is:

$$I_{coil} = \frac{5V}{70\Omega}$$

$$I_{coil} \approx 0.071A$$

$$I_{coil} \approx 71mA$$

This current level is too large to be treated as a direct comparator output load. The relay coil must be driven through the transistor or MOSFET stage.

The relay coil power is calculated by:

$$P_{coil} = V_{coil} \times I_{coil}$$

For the same example:

$$P_{coil} = 5V \times 0.071A$$

$$P_{coil} \approx 0.355W$$

The relay model should visibly change contact state when the coil is energized. If the relay coil receives the correct voltage but the contact does not change state, the selected relay model, coil rating, or wiring pins should be checked.

The flyback diode must be connected across the relay coil before the simulation is run. Its purpose is to suppress the voltage spike that occurs when the coil current is interrupted. The diode connection must satisfy this polarity:

$$Cathode \rightarrow +5V$$

$$Anode \rightarrow COIL_{LOW}$$

4.9 AC Load Simulation Setup

The load side of the simulation represents the higher power circuit controlled by the relay contacts. The alternating current source, fuse, relay common contact, normally open contact, lamp, and neutral return must be placed as a separate switching path.

The correct load current path in the relay on state is:

$$\begin{aligned} AC_{LIVE} &\rightarrow Fuse \\ &\rightarrow COM \\ &\rightarrow NO \\ &\rightarrow Lamp \\ &\rightarrow AC_{NEUTRAL} \end{aligned}$$

When the relay is off, the common contact and normally open contact are separated. Therefore, the lamp receives no complete current path and remains off.

When the relay is on, the common contact closes to the normally open contact. Therefore, the lamp receives a complete current path and turns on.

The lamp current may be estimated by:

$$I_{load} = \frac{P_{lamp}}{V_{AC}}$$

If the lamp power is (60W), and the alternating current voltage is (220V), then:

$$I_{load} = \frac{60W}{220V}$$

$$I_{load} \approx 0.273A$$

The relay contact rating must be greater than the load current. In simulation, this confirms logical function. In physical implementation, additional safety requirements are necessary, including correct insulation spacing, suitable wire rating, protective enclosure, and appropriate fuse selection.

The normally closed contact should remain unused for this project because the intended behavior is for the lamp to be off by default and on only when darkness activates the relay.

4.10 Bright State Simulation

The bright state simulation represents the condition where the light dependent resistor is exposed to sufficient light. In this condition, the resistance of the light dependent resistor is low. Because the light dependent resistor is connected from the sensing node to ground, a low resistance pulls the sensing node closer to ground.

The expected bright state condition is:

$$R_{LDR} = low$$

$$V_{SENSE} < V_{REF}$$

Since the sensing voltage is below the reference voltage, the comparator output should remain in the state that keeps the relay driver off. The MOSFET or transistor should not conduct significant relay coil current. The relay coil should remain de-energized.

The relay contact should remain in its normal position. Since the lamp is connected through the common and normally open contacts, the load circuit should remain open. Therefore, the lamp must remain off.

The bright state simulation confirms that the circuit does not activate the load when illumination is present. This is important because an uncontrolled or permanently active relay would indicate an error in the comparator input orientation, reference setting, or relay driver wiring.

The report should include the measured bright state values for V_{SENSE} , V_{REF} , $DARK_{OUT}$, V_{coil} , I_{coil} , and lamp voltage.

4.11 Dark State Simulation

The dark state simulation represents the condition where the light dependent resistor receives little or no light. In this condition, the resistance of the light dependent resistor increases. Because the light dependent resistor is connected from the sensing node to ground, an increased resistance causes the sensing voltage to rise.

The expected dark state condition is:

$$R_{LDR} = high$$

$$V_{SENSE} > V_{REF}$$

When the sensing voltage exceeds the reference voltage, the comparator output should change to the state that activates the relay driver. The driver device should conduct, and relay coil current should flow from the 5 volt supply through the relay coil and then through the driver device to ground.

The relay coil should energize, causing the relay contact to move from its normal open condition to its closed condition between common and normally open. This closes the load circuit and allows current to flow through the lamp. The lamp should turn on.

The dark state simulation confirms that the circuit performs its intended function. It proves that darkness is converted into a voltage change, the voltage change is converted into a comparator decision, the comparator decision is converted into coil current, and the coil current is converted into load switching.

The report should record the measured values of V_{SENSE} , V_{REF} , $DARK_{OUT}$, relay coil voltage, relay coil current, and lamp voltage during the dark state.

4.12 Relay Switching Verification

Relay switching verification confirms that the relay responds only when the control condition is satisfied. The relay should not switch randomly, partially, or independently of the comparator output. Its switching must be caused by relay coil energization.

The first verification step is to observe the relay coil voltage. When the relay is off, the coil voltage should be close to zero or below the relay activation requirement. When the relay is on, the coil voltage should approach the 5 volt supply value.

The second verification step is to observe the coil current. When the driver is off, the coil current should be approximately zero. When the driver is on, the coil current should match the value predicted from the coil resistance.

The third verification step is to observe the relay contacts. The common and normally open contacts should be open in the bright state and closed in the dark state. If the lamp behavior is opposite, the normally closed contact may have been used incorrectly.

The fourth verification step is to compare the relay state with the comparator output state. The relay should energize only when $DARK_{OUT}$ drives the relay driver into conduction. If the relay energizes without the expected comparator output, the driver device may be connected incorrectly.

The relay switching verification is accepted when the following logical chain is proven:

$$V_{SENSE} > V_{REF}$$

$$DARK_{OUT} = active$$

$$Driver = on$$

$$I_{coil} > 0$$

$$Relay = energized$$

$$COM \text{ connected to } NO$$

$$Lamp = on$$

4.13 Lamp On and Off Verification

The lamp on and off verification confirms that the relay contact side is correctly wired. The lamp must remain off when the relay is de energized and must turn on when the relay is energized. This test proves the correctness of the contact side of the circuit.

In the bright condition, the relay coil is not energized. The common contact is not connected to the normally open contact. Therefore, the lamp circuit is incomplete and the lamp remains off.

The expected bright state load condition is:

$$V_{lamp} \approx 0V$$

$$I_{lamp} \approx 0A$$

In the dark condition, the relay coil is energized. The common contact closes to the normally open contact. Therefore, the lamp circuit becomes complete and the lamp turns on.

The expected dark state load condition is:

$$V_{lamp} \approx V_{AC}$$

$$I_{lamp} = \frac{P_{lamp}}{V_{AC}}$$

If the lamp remains off during the dark state, the likely causes are incorrect relay contact selection, missing connection from common to the alternating current live conductor, missing connection from normally open to the lamp, missing neutral return, or a relay coil that is not energizing.

If the lamp remains on during the bright state, the likely cause is that the lamp was connected to the normally closed contact instead of the normally open contact, or that the relay driver is permanently active.

4.14 Voltage Probe Placement

Voltage probes should be placed at all nodes that determine circuit behavior. The most important voltage probe is at V_{SENSE} , because it proves that the light dependent resistor divider responds to the simulated light level.

A second voltage probe should be placed at V_{REF} , because the reference voltage determines the switching threshold. The relationship between V_{SENSE} and V_{REF} is the central decision condition of the comparator.

A third voltage probe should be placed at the comparator output node $DARK_{OUT}$. This probe confirms whether the comparator output changes state when the sensing voltage crosses the reference voltage.

A fourth voltage probe should be placed at the relay coil low side, labelled $COIL_{LOW}$.

This node shows whether the driver device is pulling the relay coil toward ground when activated.

A fifth voltage probe should be placed across the relay coil. This measurement confirms that the coil receives approximately 5 volts when the relay is intended to be energized.

A sixth voltage probe should be placed across the lamp. This measurement confirms whether the alternating current load receives voltage when the relay contact closes.

The required voltage measurements are:

$$V_{SENSE}$$

$$V_{REF}$$

$$V_{DARKOUT}$$

$$V_{COIL}$$

$$V_{LAMP}$$

These values should be recorded for both bright and dark conditions.

4.15 Current Probe Placement

Current probes should be placed in the relay coil path and the lamp path. The relay coil current confirms whether the driver stage is actually energizing the relay. The lamp current confirms whether the load side circuit is complete.

The relay coil current probe should be placed in series with the relay coil. During the bright state, the expected coil current is approximately zero. During the dark state, the expected coil current is:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

If the relay coil has a resistance of 70Ω, and the relay coil voltage is (5V), then:

$$I_{coil} \approx 71mA$$

The lamp current probe should be placed in series with the lamp. During the bright state, the expected lamp current is approximately zero. During the dark state, the lamp current depends on the lamp rating:

$$I_{lamp} = \frac{P_{lamp}}{V_{AC}}$$

For a (60W) lamp at (220V), the expected current is approximately:

$$I_{lamp} \approx 0.273A$$

The current measurements provide stronger evidence than visual lamp observation alone. A lamp symbol may appear on or off in simulation, but the measured current proves whether the load circuit is physically complete.

4.16 Waveform and Graph Observation

Waveform observation should be used to examine how the circuit changes state over time. The most important waveform comparison is between V_{SENSE} , V_{REF} , and $DARK_{OUT}$. This comparison shows whether the comparator changes output at the correct threshold.

The graph should show that V_{REF} remains approximately constant while V_{SENSE} changes with the simulated light level. When V_{SENSE} crosses above V_{REF} , the comparator output should transition into the relay activating state.

The relay coil voltage waveform should show a transition from low voltage to approximately 5 volts when the relay is activated. The coil current waveform should rise according to the relay coil path. In a simplified simulation model, the rise may appear very fast. In a more detailed inductive model, the coil current may rise gradually according to the inductive time behavior.

The basic inductive relation is:

$$v = L \frac{di}{dt}$$

This relation explains why the relay coil resists sudden current change. When the driver turns off, the coil attempts to keep current flowing. The flyback diode provides the safe path for that current.

The lamp voltage waveform should show no applied load voltage in the bright state and applied alternating current voltage in the dark state. The lamp current waveform should follow the load voltage when the relay contact is closed.

Waveform observation is important because it reveals hidden errors that may not be visible from the static schematic. It can show floating comparator outputs, unstable threshold switching, relay chatter, missing flyback protection, and incorrect load side connections.

4.17 Expected Simulation Results

The expected simulation results must follow the designed cause and effect sequence. In the bright state, the light dependent resistor has low resistance, the sensing voltage remains below the reference voltage, the comparator output keeps the driver off, the relay coil is not energized, the common and normally open contacts remain open, and the lamp remains off.

In the dark state, the light dependent resistor has high resistance, the sensing voltage rises above the reference voltage, the comparator output activates the driver, the relay coil becomes energized, the common and normally open contacts close, and the lamp turns on.

The expected mathematical conditions are:

Bright state:

$$R_{LDR} = low$$

$$V_{SENSE} < V_{REF}$$

$$I_{coil} \approx 0A$$

$$V_{lamp} \approx 0V$$

$$Lamp = off$$

Dark state:

$$R_{LDR} = high$$

$$V_{SENSE} > V_{REF}$$

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

$$V_{lamp} \approx V_{AC}$$

$$Lamp = on$$

The flyback diode should remain reverse biased during normal relay activation.

During relay turn off, it should conduct only long enough to dissipate the stored magnetic energy of the coil. If the diode is missing, the relay turn off event may produce a voltage spike or simulation instability.

The expected result is therefore not only that the lamp turns on in darkness. The complete expected result is that every stage performs its correct role in the correct order, with measurable voltage and current values supporting the observed behavior.

4.18 Simulation Result Table

The simulation results should be recorded in a structured table so that the bright and dark states can be compared clearly. The table must include the input condition, sensing voltage, reference voltage, comparator output, driver state, relay state, lamp state, and interpretation.

A suitable result table is shown below.

Table 4.1: Expected simulation result table.

Test condition	Expected R_{LDR}	Expected V_{SENSE}	Comparator condition	Driver state	Relay state	Lamp state	Result interpretation
Bright condition	Low	Less than V_{REF}	$V_{SENSE} < V_{REF}$	Off	De energized	Off	Correct if lamp remains off
Dark condition	High	Greater than V_{REF}	$V_{SENSE} > V_{REF}$	On	Energized	On	Correct if lamp turns on

A more detailed measurement table should also be included.

Table 4.3: Detailed simulation measurement table.

Measurement point	Bright state expected value	Dark state expected value	Purpose
V_{SENSE}	Below V_{REF}	Above V_{REF}	Confirms LDR divider behavior
V_{REF}	Constant threshold value	Constant threshold value	Confirms reference stability
$DARK_{OUT}$	Driver inactive level	Driver active level	Confirms comparator switching
V_{coil}	Approximately (0V)	Approximately (5V)	Confirms relay coil activation
I_{coil}	Approximately (0A)	Calculated coil current	Confirms relay energization
V_{lamp}	Approximately (0V)	Approximately V_{AC}	Confirms load switching
I_{lamp}	Approximately (0A)	Calculated load current	Confirms complete load path

The final report should include actual measured values from Proteus in the table. If the measured values differ from the expected values, the difference should be explained rather than ignored.

4.19 Simulation Acceptance Criteria

The simulation should be accepted only if it satisfies all functional, electrical, and protection requirements. The first acceptance criterion is that the light dependent resistor divider must produce a measurable change in V_{SENSE} when the simulated light level is changed.

The second acceptance criterion is that the comparator must switch state when V_{SENSE} crosses V_{REF} . The comparator output must not remain floating, unstable, or permanently locked in one state.

The third acceptance criterion is that the relay driver must energize the relay coil only in the dark condition. The relay coil current must be close to the value predicted by the coil voltage and coil resistance.

The fourth acceptance criterion is that the relay contact must switch the load correctly. The lamp must be off in the bright condition and on in the dark condition. This confirms that the common and normally open contacts are correctly used.

The fifth acceptance criterion is that the flyback diode must be correctly connected across the relay coil. Its cathode must be connected to the positive coil side, and its anode must be connected to the driver side of the coil.

The sixth acceptance criterion is that the low voltage control side and the alternating current load side must remain electrically separated except through the relay mechanism. There must be no direct connection between the alternating current load conductor and the 5 volt control ground.

The seventh acceptance criterion is that all measured values must agree with the theoretical expectations within reasonable simulation tolerance. The required acceptance conditions can be stated as follows:

$$\text{Bright condition : } V_{SENSE} < V_{REF}, \quad I_{coil} \approx 0A, \quad Lamp = off$$

$$\text{Dark condition : } V_{SENSE} > V_{REF}, \quad I_{coil} > 0A, \quad Lamp = on$$

If any one of these conditions is not satisfied, the simulation should not be considered successful. The circuit should then be corrected in the troubleshooting chapter.

4.20 Chapter Summary

This chapter established the simulation procedure for the light activated relay power driver circuit. The Proteus simulation was structured to verify every major part of the design, including the light dependent resistor divider, reference voltage, comparator output, relay driver stage, relay coil, flyback diode, relay contacts, and alternating current lamp circuit.

The simulation was divided into two main operating states. In the bright state, the light dependent resistor resistance is low, the sensing voltage remains below the reference voltage, the relay driver remains inactive, the relay coil is not energized, and the lamp remains off. In the dark state, the light dependent resistor resistance increases, the sensing voltage rises above the reference voltage, the relay driver activates, the relay coil energizes, the relay contact closes, and the lamp turns on.

The chapter also defined the required voltage probes, current probes, waveform observations, result tables, and acceptance criteria. These verification steps ensure that the simulation is not judged only by visual lamp behavior, but by measurable electrical evidence. A successful simulation must show correct sensing, correct threshold com-

parison, correct relay driving, correct load switching, correct flyback protection, and correct separation between the low voltage control circuit and the higher voltage load circuit.

Therefore, Chapter 4 proves the operational validity of the designed circuit and prepares the foundation for Chapter 5, where possible faults and their corrections are analyzed systematically.

5

Troubleshooting

5.1 Introduction to Troubleshooting

Troubleshooting is the systematic process of identifying, isolating, explaining, and correcting faults that prevent an electrical circuit from operating according to its intended design. In this project, troubleshooting is especially important because the circuit contains both a low voltage control section and a higher power load switching section. The control section includes the light dependent resistor, the reference voltage network, the comparator, the output pull up resistor, the transistor or metal oxide semiconductor field effect transistor driver, the relay coil, and the flyback diode. The load section includes the relay contacts, the alternating current supply, the fuse, and the lamp.

05 — TROUBLESHOOTING: why Proteus crashes or spikes when the relay turns OFF

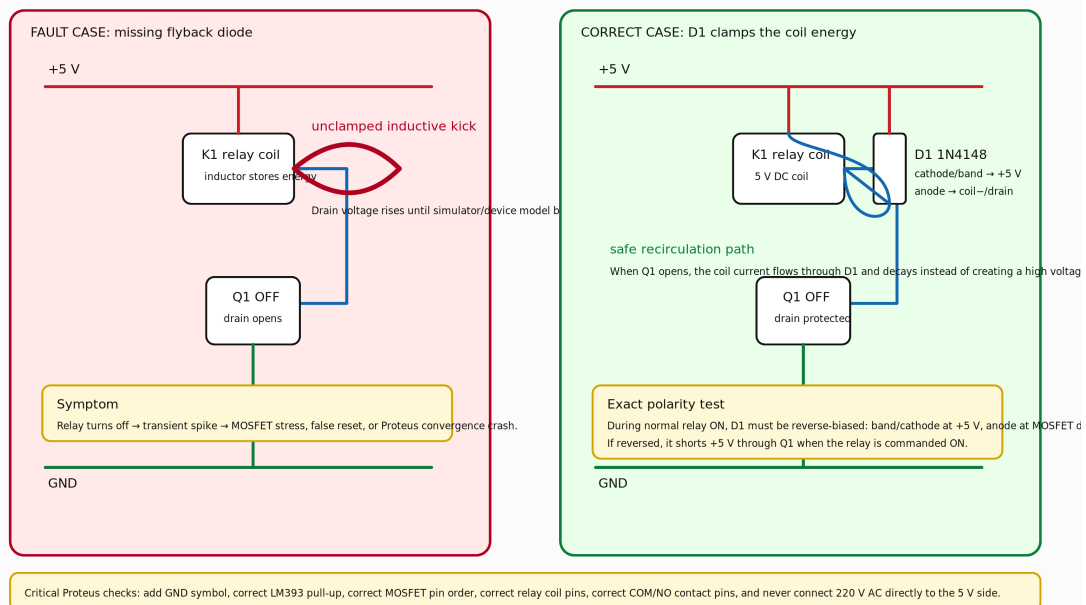


Figure 5.1: Troubleshooting flyback and polarity Proteus style.

The purpose of this chapter is to define the most likely faults that may occur during

the Proteus simulation of the relay power driver circuit and to provide exact diagnostic procedures for each fault. A correct troubleshooting method must not rely on guessing. Each fault must be examined according to measurable electrical evidence, such as voltage level, current flow, node state, contact state, and component polarity.

The relay circuit must be analyzed as a chain of dependent stages. The sensing stage must produce a valid sensor voltage. The comparator stage must compare this sensor voltage with a reference voltage. The output stage must produce a valid control signal. The driver stage must supply enough coil current. The relay coil must energize correctly. The contact side must close the correct path to the load. The flyback diode must suppress the inductive voltage spike when the relay is switched off. If one stage fails, every later stage may also appear to fail, even if those later components are not damaged or incorrectly connected.

For this reason, the troubleshooting method used in this chapter follows a strict order. The power supply is checked first, followed by the sensor voltage, reference voltage, comparator output, driver input, relay coil voltage, relay contact continuity, load voltage, and protection components. This order prevents false conclusions and allows the exact cause of the fault to be located with minimum uncertainty.

5.2 Relay Does Not Click

A relay click is the mechanical indication that the relay coil has produced enough magnetic force to move the internal contact armature. If the relay does not click during the dark condition, the fault must be searched on the coil control side before the load side is blamed. The lamp may remain off because the relay contact never changes position, but the primary problem may still be in the sensor, comparator, driver, or coil wiring.

The first point to verify is the voltage across the relay coil. The coil should receive approximately 5 volts when the circuit is in the dark activated state. The voltage across the relay coil may be written as:

$$V_{coil} = V_{CC} - V_{DS}$$

where V_{CC} is the 5 volt supply and V_{DS} is the voltage across the switching device when it is on. If the driver device is operating correctly, V_{DS} should be small, and most of the supply voltage should appear across the relay coil.

The expected relay coil current is calculated by Ohm's law:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

If the relay coil resistance is 70Ω and the coil voltage is (5V), then:

$$I_{coil} = \frac{5}{70} = 0.071A$$

Therefore, the relay requires approximately (71mA). If the driver cannot provide this current, the relay will not energize.

If the relay does not click, the following checks must be made in order. Confirm that

one side of the relay coil is connected to (+5V). Confirm that the other side of the relay coil is connected to the drain or collector of the driver device. Confirm that the source or emitter of the driver device is connected to ground. Confirm that the comparator output is producing the correct activation signal. Confirm that the driver gate or base receives a valid switching voltage. Confirm that the relay model in Proteus is a 5 volt relay and not a 12 volt or 24 volt relay.

A common error is attempting to connect the comparator output directly to the relay coil. This is not correct because the comparator output is a signal node, not a coil power source. The relay coil requires a driver stage capable of carrying the coil current. Therefore, if the relay does not click and the comparator output appears correct, the driver stage must be examined immediately.

5.3 Lamp Does Not Turn ON

If the relay clicks but the lamp does not turn on, the control side is likely functioning, and the fault is probably located on the contact side of the relay or in the load circuit. The relay click proves that the coil is energized, but it does not prove that the correct contacts are wired to the lamp.

The lamp should be connected through the normally open contact if the desired behavior is that the lamp turns on only when darkness activates the relay. The correct current path is:

AC Live → *Fuse*
 → *COM*
 → *NO*
 → *Lamp*
 → *AC Neutral*

When the relay is off, the common contact and normally open contact are separated, so the lamp remains off. When the relay is energized, the common contact connects to the normally open contact, completing the load circuit and turning the lamp on.

If the lamp does not turn on although the relay clicks, the voltage across the lamp must be measured. If the measured lamp voltage is approximately equal to the supply voltage, the lamp should illuminate. If the lamp voltage is zero or very small, the contact path is incomplete. The most likely causes are wrong relay contact selection, open connection between common and normally open, missing alternating current source, incorrectly connected lamp, missing neutral return, or an incorrectly placed fuse.

The lamp current can be estimated by:

$$I_{load} = \frac{P_{lamp}}{V_{AC}}$$

For a (60W) lamp connected to (220V):

$$I_{load} = \frac{60}{220} = 0.273A$$

This confirms that the relay contact must be rated above the load current. In simulation, the lamp may still operate even if realistic safety margins are not modeled, but an academic report must state that real hardware requires correct contact rating, fuse selection, insulation spacing, and enclosure protection.

5.4 Lamp Always Remains ON

If the lamp always remains on, even when the relay is not activated, the most probable cause is that the lamp is connected through the normally closed contact instead of the normally open contact. In a single pole double throw relay, the common contact is connected to the normally closed contact when the coil is not energized. Therefore, if the lamp is connected through common and normally closed, the lamp will naturally remain on during the inactive state.

The expected dark activated behavior requires the common and normally open contacts. The normally closed contact must remain unused unless the intended function is inverse operation. If the lamp must turn on only when darkness is detected, the connection must be:

$$COM \rightarrow NO$$

not:

$$COM \rightarrow NC$$

Another possible cause is that the relay coil is always energized. This may happen if the comparator inputs are reversed, the threshold is incorrectly set, the driver gate is permanently high, or the transistor is wired incorrectly. Therefore, the troubleshooting process must determine whether the lamp is always on because of wrong contact wiring or because the relay is always activated.

The distinction is simple. If the relay is not clicking and the lamp is still on, the contact wiring is probably using the normally closed contact. If the relay is energized continuously and the lamp is on, the fault is probably in the comparator, threshold, driver, or sensor stage.

5.5 Relay Always Remains ON

If the relay always remains on, the coil is continuously receiving current. This fault means that the driver device is continuously conducting or that the relay coil has been connected directly to ground without proper switching control.

The first measurement should be taken at the driver input. For a metal oxide semiconductor field effect transistor, the gate voltage must be measured with respect to ground. If the gate voltage remains high in both light and dark conditions, the com-

parator output or pull up network is forcing the driver on. If the gate voltage is low but the relay remains energized, the driver device may be incorrectly connected, internally shorted in the simulation model, or replaced with the wrong component type.

The comparator condition must also be checked. In the selected circuit orientation, the sensor voltage rises as the light dependent resistor becomes darker. The relay should activate when:

$$V_{sense} > V_{ref}$$

If the comparator is wired so that the non inverting and inverting inputs are exchanged, the output behavior may become inverted. The relay may then activate in light instead of darkness, or it may remain active because the reference setting places the circuit permanently in the active region.

The potentiometer reference voltage must also be examined. If V_{ref} is set too low, then V_{sense} may remain greater than V_{ref} under both light and dark conditions. In that case, the comparator sees the activation condition all the time. The threshold should be adjusted so that:

$$V_{sense,light} < V_{ref} < V_{sense,dark}$$

This inequality is essential. It means the reference voltage must be placed between the expected sensor voltage in light and the expected sensor voltage in darkness.

5.6 Relay Chatters or Switches Unstably

Relay chatter means that the relay rapidly switches on and off instead of changing state cleanly. This usually occurs when the comparator input voltage is very close to the reference voltage. Small variations in the light dependent resistor, supply voltage, numerical simulation, or noise may cause the comparator output to alternate between high and low.

The unstable condition occurs when:

$$V_{sense} \approx V_{ref}$$

In this condition, even a very small voltage disturbance can change the comparator output state. Since the relay is mechanical, repeated fast switching produces audible clicking and unstable lamp operation.

The first correction is to adjust the potentiometer so that the threshold is clearly separated from the normal operating sensor voltage. The desired condition for reliable switching is:

$$V_{sense,light} < V_{ref}$$

and:

$$V_{sense,dark} > V_{ref}$$

with enough margin between these values. The margin can be expressed as:

$$M_{light} = V_{ref} - V_{sense,light}$$

$$M_{dark} = V_{sense,dark} - V_{ref}$$

Both margins should be positive. Larger margins produce more stable switching.

A stronger correction is to add hysteresis to the comparator circuit. Hysteresis creates two different switching thresholds, one for turning on and one for turning off. This prevents the relay from rapidly toggling near a single threshold point. Although hysteresis may not be required for the basic project, it is an academically important improvement because real light sensor circuits are often exposed to gradual light changes and electrical noise.

5.7 Comparator Output Does Not Change

If the comparator output does not change when the light dependent resistor changes from light to dark, the comparator stage must be examined independently from the relay and load. The comparator output depends on three essential conditions: correct supply voltage, valid input voltages, and correct output pull up connection.

The supply pins of the comparator must be connected correctly. The positive supply pin must be connected to (+5V), and the ground pin must be connected to the control ground. If the comparator has no supply voltage, its output cannot represent the comparison result.

The input voltages must then be measured. The voltage at the sensor node must change when the light dependent resistor changes. If V_{sense} does not change, the fault is in the light dependent resistor divider, not in the comparator. The reference voltage must also be stable. If V_{ref} is floating, incorrectly connected, or equal to an extreme value near (0V) or (5V), the comparator may remain permanently in one output state.

The comparator output condition is determined by:

$$V_{out} = High \quad when \quad V_+ > V_-$$

and:

$$V_{out} = Low \quad when \quad V_+ < V_-$$

In the intended design, V_+ is connected to V_{sense} , and V_- is connected to V_{ref} . Therefore, the output should change when V_{sense} crosses V_{ref} .

If all input voltages are correct but the output still does not change, the pull up resistor must be checked. A comparator such as the LM393 uses an output structure that requires an external resistor to create a valid high level. Without this resistor, the

output may not rise correctly, even when the internal transistor is off.

5.8 Missing Pull Up Resistor Problem

The missing pull up resistor is one of the most important hidden faults in this circuit. The comparator output cannot be treated as a normal logic output unless the device model provides an internal high side drive. In the selected comparator configuration, the output requires an external resistor connected from the output node to (+5V).

The correct connection is:

$$+5V \rightarrow 10k\Omega \rightarrow \text{Comparator Output}$$

The output node is then connected to the driver input through the gate or base resistor.

If this resistor is missing, the output node may float when the comparator output transistor is off. A floating node has no defined voltage. It may appear high, low, unstable, or dependent on simulation conditions. This can cause the relay not to click, to chatter, or to behave differently each time the simulation starts.

The pull up resistor current when the comparator output is low can be estimated by:

$$I_{pullup} = \frac{V_{CC}}{R_{pullup}}$$

$$\text{For}(V_{CC} = 5V) \text{ and } (R_{pullup} = 10k\Omega) :$$

$$I_{pullup} = \frac{5}{10000} = 0.0005A$$

$$I_{pullup} = 0.5mA$$

This current is small enough for the comparator output to sink safely in simulation while still producing a valid high level when the output transistor is off.

5.9 Incorrect Comparator Input Polarity

Incorrect comparator input polarity means that the sensor voltage and reference voltage are connected to the wrong comparator inputs for the intended behavior. A comparator does not know the meaning of light or darkness. It only compares the voltage at the non inverting input with the voltage at the inverting input.

The intended operation is darkness activation. In the selected voltage divider orientation, the sensor voltage rises when darkness increases. Therefore, the relay should turn on when:

$$V_{sense} > V_{ref}$$

To obtain this behavior, the non inverting input must receive V_{sense} , and the inverting input must receive V_{ref} . The correct connection is:

$$IN+ \rightarrow V_{sense}$$

$$IN- \rightarrow V_{ref}$$

If these two inputs are exchanged, the output becomes inverted. The relay may activate in bright light and deactivate in darkness. This is not a random fault. It is a direct logical consequence of reversing the comparison condition.

The correction is to verify the comparator input pins in the Proteus model and confirm that the circuit symbol matches the actual pin function. The labels (+) and (-) must be trusted more than the visual position of the pins. Some schematic symbols place pins differently from the physical package, so the function label is more important than the drawing layout.

5.10 Incorrect MOSFET or Transistor Wiring

The driver device is responsible for allowing relay coil current to flow. If this component is wired incorrectly, the comparator may operate correctly while the relay still fails. The driver must be placed as a low side switch in the recommended design.

For a metal oxide semiconductor field effect transistor, the correct wiring is:

$$Drain \rightarrow Relay\ Coil\ Low\ Side$$

$$Source \rightarrow Ground$$

$$Gate \rightarrow Comparator\ Output\ Through\ Gate\ Resistor$$

The relay coil high side must connect to (+5V). When the gate voltage is high enough relative to the source, the transistor conducts and the coil current flows through the drain and source path.

For a bipolar junction transistor, the corresponding structure is:

$$Collector \rightarrow Relay\ Coil\ Low\ Side$$

$$Emitter \rightarrow Ground$$

$$Base \rightarrow Comparator\ Output\ Through\ Base\ Resistor$$

A common error is reversing drain and source, or collector and emitter. Another common error is connecting the relay coil to the gate or base instead of the drain or

collector. The gate or base is a control terminal, not the coil current path.

For a bipolar transistor, the base current must be sufficient to saturate the transistor. If the relay coil current is (71mA) and a forced current gain of 10 is used for saturation, the required base current is:

$$I_B = \frac{I_{coil}}{10}$$

$$I_B = \frac{71mA}{10} = 7.1mA$$

The base resistor is then estimated by:

$$R_B = \frac{V_{out} - V_{BE}}{I_B}$$

If ($V_{out} = 5V$), ($V_{BE} = 0.7V$), and ($I_B = 7.1mA$), then :

$$R_B = \frac{4.3}{0.0071} = 606\Omega$$

A practical value near 560Ω or 680Ω may be used in simulation. If a metal oxide semiconductor field effect transistor is used instead, the gate current is not calculated in the same way because the gate is voltage controlled. However, a small gate resistor and a large pull down resistor are still recommended for stable switching.

5.11 Relay Coil Wiring Error

Relay coil wiring errors occur when the coil terminals are confused with the contact terminals, when the coil does not receive the correct voltage, or when the coil is not placed in the driver current path. The relay contains two different electrical systems: the coil system and the contact system. The coil belongs to the 5 volt control side. The contacts belong to the load switching side.

The correct coil current path is:

$$+5V \rightarrow \text{Relay Coil} \rightarrow \text{Driver Device} \rightarrow \text{Ground}$$

If this path is open at any point, the relay will not click. If the coil is connected directly between (+5V) and ground without the driver device, the relay will remain permanently energized. If the coil is connected to the alternating current load side, the circuit becomes incorrect and unsafe.

The relay coil resistance can be measured or checked from the component properties. The expected current is:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

If R_{coil} is too low, the driver may be overloaded. If R_{coil} is too high for the selected

relay model, the coil current may be too low to activate the relay. If the relay is rated for 12 volts but the circuit provides only 5 volts, the relay may not energize.

The correction is to select a 5 volt relay model, connect one coil terminal to (+5V), connect the other coil terminal to the driver drain or collector, and keep the contact pins separate from the coil pins.

5.12 Flyback Diode Missing

The flyback diode is required because the relay coil stores magnetic energy while current flows through it. When the driver turns off, the coil current cannot instantly become zero. The collapsing magnetic field generates a voltage spike that attempts to keep current flowing. Without a safe path, this voltage may rise high enough to disturb the simulation, damage the driver model, or create unrealistic numerical instability.

The energy stored in the relay coil is expressed as:

$$E = \frac{1}{2}LI^2$$

where (L) is the coil inductance and I is the coil current before turn off. This equation shows that the coil contains stored energy whenever current flows. That energy must go somewhere when the relay is switched off.

The flyback diode provides the safe path. It is connected across the relay coil in reverse bias during normal operation. When the coil voltage reverses during turn off, the diode becomes forward biased and allows the coil current to decay gradually.

If the diode is missing, the most common symptoms are simulation crash, voltage spikes at the driver drain or collector, relay turn off instability, and possible failure of the driver component in a more realistic model. The correction is to place a 1N4148 diode directly across the relay coil with correct polarity.

5.13 Flyback Diode Reversed

A reversed flyback diode is not equivalent to a missing diode. It creates a different and more severe fault. If the diode is connected in the forward direction across the coil during normal operation, it creates a near short circuit across the supply when the driver turns on. This may prevent the relay from energizing, overload the driver, collapse the supply voltage, or cause immediate simulation failure.

The correct flyback diode orientation is:

$$\text{Cathode} \rightarrow +5V$$

$$\text{Anode} \rightarrow \text{Relay Coil Low Side}$$

The cathode is usually marked by a band on the diode symbol or package. The anode must connect to the driver side of the coil. In normal operation, the diode is reverse biased and does not conduct. During turn off, the coil reverses its voltage, and the

diode becomes forward biased.

The normal state condition is:

$$V_{cathode} > V_{anode}$$

so the diode remains off. The turn off condition reverses the coil voltage and allows the diode to conduct. If this polarity is reversed, the diode conducts during normal operation, which is incorrect.

The correction is to inspect the diode symbol carefully and ensure that the banded side is connected to the positive side of the relay coil.

5.14 Wrong Relay Contact Selection

Wrong relay contact selection occurs when the load is connected to the wrong pair of relay contacts. A single pole double throw relay usually contains common, normally open, and normally closed terminals. The common terminal moves between the normally closed terminal and the normally open terminal depending on coil state.

For this project, the required behavior is that the lamp turns on when darkness activates the relay. Therefore, the lamp must be connected through common and normally open. The correct load path is:

$$\begin{aligned} &Live \rightarrow Fuse \\ &\quad \rightarrow COM \\ &\quad \rightarrow NO \\ &\quad \rightarrow Lamp \\ &\quad \rightarrow Neutral \end{aligned}$$

If common and normally closed are used, the behavior is inverted. The lamp will be on when the relay is off and may turn off when the relay energizes. This is a wiring logic error, not a sensor or comparator error.

The correction is to identify the relay contact labels in Proteus and ensure that common is connected to the supply side of the load path while normally open is connected to the lamp side. The normally closed terminal must remain unused unless inverse operation is intentionally required.

5.15 AC Load Wiring Error

The alternating current load side must form a complete circuit through the relay contacts and the lamp. A lamp cannot turn on unless there is a complete path from live to neutral through the load. The relay contact is only a switch in this path. It does not create voltage by itself.

The correct load circuit is:

AC Live → *Fuse*
 → *Relay COM*
 → *Relay NO*
 → *Lamp*
 → *AC Neutral*

The most common load wiring errors are connecting both lamp terminals to the same side of the supply, omitting the neutral return, connecting the lamp directly across the source while bypassing the relay, connecting the relay contacts in parallel with the lamp, or placing the fuse in a position that does not protect the live conductor.

The voltage across the lamp must be measured during relay activation. If:

$$V_{lamp} \approx V_{AC}$$

then the lamp should turn on. If:

$$V_{lamp} \approx 0$$

then the load path is open or incorrectly wired.

In a real circuit, the live conductor should be switched rather than the neutral conductor, because switching the live conductor reduces the chance that the load remains internally energized when it appears off. In simulation, the lamp may work either way, but an academic report must describe the safer wiring discipline.

5.16 Control Side and Load Side Isolation Error

The control side and load side must remain electrically separated. The control side uses 5 volt direct current. The load side may use a much higher alternating current voltage. The relay allows control transfer without a direct electrical connection between these two domains. This separation is one of the main reasons for using a relay.

An isolation error occurs when the ground of the 5 volt control side is directly connected to the alternating current load side, when the alternating current live conductor is accidentally connected to a control node, or when relay coil pins are confused with contact pins. Such an error destroys the intended separation of the system.

The correct separation rule is:

Control Side ≠ *Load Side*

The only intended coupling is the internal electromechanical action of the relay. The relay coil is energized by the 5 volt circuit. The relay contact mechanically changes state and switches the load circuit. No conductor should directly connect (+5V), control ground, sensor nodes, comparator pins, or driver pins to the alternating current live or neutral conductors.

In Proteus, an isolation error may produce unrealistic results, simulation warnings,

short circuits, or dangerous looking voltage conditions. In real hardware, this type of error can be life threatening. Therefore, the academic report must state that the simulation demonstrates the principle only and that real mains circuits require certified design practices, rated components, protective enclosure, insulation distance, and qualified supervision.

5.17 Simulation Crash During Relay Turn Off

A simulation crash during relay turn off is commonly caused by an unmanaged inductive voltage spike. The relay coil is an inductor, and the inductor current cannot stop instantly. When the driver opens the coil path, the simulator attempts to solve a very rapid voltage transition. If no flyback path exists, the voltage may rise to an extreme value, causing numerical convergence failure.

The governing inductor relation is:

$$V_L = L \frac{dI}{dt}$$

This equation shows that when the current changes very rapidly, the inductor voltage becomes large. During sudden turn off, $\frac{dI}{dt}$ may become very large in magnitude, so the coil voltage can also become very large.

The correction is to add the flyback diode across the relay coil. The diode reduces the voltage stress by allowing current to circulate after the driver turns off. The current then decays more slowly and the simulation becomes stable.

A secondary correction is to ensure that all reference nodes are properly grounded on the control side. Floating nodes can also cause convergence problems. However, if the crash occurs specifically when the relay turns off, the missing or reversed flyback diode is the most probable cause.

5.18 Systematic Debugging Procedure

A systematic debugging procedure must follow the signal path from input to output. The circuit should not be debugged randomly, because random debugging can produce false conclusions. The correct method is to verify each stage before moving to the next stage.

First, verify the 5 volt supply. The comparator, divider, relay coil, and driver all depend on this supply. If the supply is incorrect, all later measurements are unreliable.

Second, verify the sensor voltage. Measure V_{sense} under light and dark conditions. It must change when the light dependent resistor condition changes. If V_{sense} does not change, inspect the light dependent resistor, the fixed resistor, and the divider wiring.

Third, verify the reference voltage. Measure V_{ref} at the potentiometer wiper. The reference must be between the light state sensor voltage and the dark state sensor voltage:

$$V_{sense,light} < V_{ref} < V_{sense,dark}$$

Fourth, verify the comparator output. The output should change state when V_{sense} crosses V_{ref} . If it does not, inspect the comparator supply pins, input polarity, and output pull up resistor.

Fifth, verify the driver input and output. The gate or base must receive a valid activation signal. The driver output terminal must switch the relay coil low side toward ground.

Sixth, verify relay coil voltage and current. The coil must receive sufficient voltage and current to energize.

Seventh, verify the relay contact path. Confirm that common and normally open are used for dark activated lamp operation.

Eighth, verify the load voltage. The lamp must receive the alternating current supply voltage when the relay is energized.

Ninth, verify the flyback diode. Confirm that the cathode is connected to the positive coil side and the anode is connected to the driver side of the coil.

This order creates a complete diagnostic chain from cause to effect.

5.19 Troubleshooting Table

Table 5.1: Troubleshooting table.

Fault condition	Probable cause	Measurement to perform	Corrective action
Relay does not click	No coil voltage, weak driver, wrong relay rating, or missing control signal	Measure V_{coil} , I_{coil} , and driver input voltage	Use a 5 volt relay, correct the driver wiring, and verify comparator output
Lamp does not turn on	Relay contact path is open or wrong contact is used	Measure voltage across the lamp during relay activation	Connect the load through common and normally open
Lamp always remains on	Load connected through normally closed contact or relay always energized	Observe relay state and inspect contact labels	Use common and normally open for dark activated operation
Relay always remains on	Comparator output permanently active or driver permanently conducting	Measure V_{sense} , V_{ref} , and driver gate or base voltage	Adjust threshold, correct comparator input polarity, and inspect driver wiring
Relay chatters	Sensor voltage is too close to reference voltage	Compare V_{sense} and V_{ref} during switching	Adjust the threshold or add hysteresis

Fault condition	Probable cause	Measurement to perform	Corrective action
Comparator output does not change	No supply, wrong input polarity, fixed sensor voltage, or missing pull up resistor	Measure comparator supply, inputs, and output	Correct supply wiring, input wiring, divider wiring, and pull up resistor
Output node floats	Missing pull up resistor	Measure output voltage when comparator should be high	Add a 10 kilo ohm pull up resistor to 5 volts
Relay coil overheats or circuit current is excessive	Reversed flyback diode or incorrect coil connection	Inspect diode polarity and coil path	Connect diode cathode to 5 volts and anode to coil low side
Simulation crashes at turn off	Missing flyback path for inductive energy	Observe voltage spike at driver drain or collector	Add correctly polarized flyback diode
Lamp receives no voltage	Alternating current path is incomplete	Measure V_{lamp} during relay activation	Complete the live to fuse to common to normally open to lamp to neutral path
Unsafe domain connection	Control side connected directly to load side	Inspect all wires between 5 volt section and alternating current section	Remove direct connection and keep only relay contact switching

5.20 Chapter Summary

This chapter presented a complete troubleshooting framework for the light activated relay power driver circuit. The analysis showed that faults must be diagnosed according to the internal order of the system, beginning with the supply and ending with the load. The relay does not operate independently. It depends on correct sensor voltage, correct reference voltage, correct comparator operation, correct output pull up, correct driver wiring, correct coil current, correct flyback protection, and correct contact side load wiring.

The most important conclusion is that the circuit must be treated as a sequence of linked electrical conditions. The sensor must generate a changing voltage. The comparator must compare that voltage against a valid reference. The comparator output must be pulled up to a defined high level. The driver must supply the relay coil current. The relay coil must be protected by a correctly polarized flyback diode. The contacts must switch the load through common and normally open terminals. The control side and load side must remain separated.

A fault in this circuit can therefore be solved only by locating the exact failed stage. If the relay does not click, the coil and driver path must be inspected. If the relay clicks but the lamp does not turn on, the contact and load path must be inspected. If the relay

chatters, the threshold region must be inspected. If the simulation crashes during turn off, the flyback diode must be inspected. If the system behaves opposite to the intended design, the comparator input polarity or relay contact selection must be inspected.

The chapter establishes that reliable troubleshooting is not a secondary activity after design. It is part of the design proof itself. A circuit that cannot be debugged clearly is not yet fully understood. This relay power driver becomes academically complete only when its normal operation, failure modes, diagnostic measurements, and correction methods are all explicitly explained.

6

Questions

6.1 Introduction to Review Questions

This chapter presents the formal review questions for the light-activated relay power driver project. The purpose of this chapter is to verify that the theoretical explanation, calculation process, circuit scheme, Proteus simulation, troubleshooting method, and safety principles have been understood with sufficient technical accuracy.

The questions in this chapter are organized according to the structure of the report. The first group examines the basic concept of using a low-power control circuit to switch a higher-power load. The following groups examine the relay, the light-dependent resistor, the comparator, the relay driver, the flyback diode, the mathematical calculations, the Proteus schematic, the simulation behavior, troubleshooting procedures, and electrical safety. The final groups prepare the student for academic discussion and oral defense.

The chapter is not intended to repeat the entire report. Its function is to test the correctness of the design from multiple technical directions. A correct answer must identify the component involved, explain its role in the system, and connect that role to the complete operation of the circuit. A weak answer only describes a component in isolation. A strong answer explains how that component contributes to the full sequence of detection, decision, driving, switching, protection, and load activation.

The central operating principle of the project is that the light-dependent resistor detects the light condition, the comparator determines whether the sensed voltage has crossed the selected threshold, the relay driver supplies sufficient coil current, the relay provides contact-based switching, and the flyback diode protects the switching device from the reverse voltage generated by the relay coil when it is turned off.

6.2 Basic Concept Questions

6.2.1 What is the main purpose of this project?

The main purpose of this project is to design and simulate a circuit in which a low-power light-sensing control stage can switch a higher-power electrical load through a relay. The circuit demonstrates how a small control signal can operate a load that requires more current or a different voltage domain than the sensing circuit can safely provide directly.

6.2.2 Why is the project called a power driver?

The project is called a power driver because the sensing circuit does not directly supply the load current. Instead, the sensing circuit produces a control signal, and this control signal operates a driver stage. The driver stage supplies the relay coil current, while the relay contacts switch the higher-power load circuit.

6.2.3 What is meant by a low-power control circuit?

A low-power control circuit is the part of the system that uses a small voltage and small current to process information or produce a control decision. In this project, the light-dependent resistor, the reference voltage circuit, the comparator, and the gate control path belong to the low-power control side.

6.2.4 What is meant by a high-power load circuit?

A high-power load circuit is the part of the system that requires greater electrical power than the sensor and comparator circuit can provide directly. In this project, the lamp or motor load represents the higher-power side. The load side may use alternating current, while the control side uses a low-voltage direct current supply.

6.2.5 Why should the control circuit not be connected directly to the load?

The control circuit should not be connected directly to the load because the sensing and comparator components are not designed to carry high load current. Direct connection would overload the control components, cause malfunction, damage the circuit, and may remove the required separation between the low-voltage side and the high-voltage side.

6.2.6 What is the complete operating sequence of the circuit?

The operating sequence begins when the light level around the light-dependent resistor changes. This change modifies the resistance of the light-dependent resistor and therefore changes the sensed voltage. The comparator compares the sensed voltage with the reference voltage. If the sensed voltage reaches the required condition, the comparator output changes state. The driver stage then energizes the relay coil. The

relay contact moves from its resting condition to its activated condition. The load circuit becomes complete, and the lamp turns on.

6.3 Relay Operation Questions

6.3.1 What is a relay?

A relay is an electromechanical switching device. It contains a coil and one or more contacts. When current flows through the coil, a magnetic field is produced. This magnetic field moves the internal contact mechanism and changes the connection state of the relay contacts.

6.3.2 Why is a relay used in this project?

A relay is used because it allows the low-voltage control side to switch a separate load circuit without carrying the full load current through the control components. The relay coil belongs to the control side, while the relay contacts belong to the load side.

6.3.3 What happens when the relay coil is energized?

When the relay coil is energized, current flows through the coil and creates a magnetic field. This magnetic field pulls the internal movable contact from its normal position to its activated position. In a normally open contact path, the contact closes and allows current to flow through the load.

6.3.4 What is the difference between normally open and normally closed relay contacts?

A normally open contact is open when the relay coil is not energized. It closes only when the relay coil is energized. A normally closed contact is closed when the relay coil is not energized. It opens when the relay coil is energized.

6.3.5 Which relay contact should be used if the lamp must turn on only in darkness?

The common contact and the normally open contact should be used. In this arrangement, the lamp remains off when the relay is not energized. When darkness causes the relay coil to energize, the common contact connects to the normally open contact, and the lamp turns on.

6.3.6 What would happen if the normally closed contact were used instead?

If the normally closed contact were used, the lamp would be on when the relay coil is not energized. When the relay activates, the normally closed contact would open, and the lamp would turn off. This behavior is the opposite of the required dark-activated lamp operation.

6.3.7 Why does the relay provide separation between the control side and the load side?

The relay provides separation because the coil and the contacts are not the same electrical conductor. The control side energizes the coil, and the magnetic action of the coil moves the contacts. The load circuit is switched through the contacts, not through the comparator or sensor wires.

6.4 LDR Sensor Questions

6.4.1 What is a light-dependent resistor?

A light-dependent resistor is a resistor whose resistance changes according to the light intensity falling on its surface. Its resistance is lower in bright light and higher in darkness.

6.4.2 What is the function of the light-dependent resistor in this project?

The function of the light-dependent resistor is to sense the light level and convert the physical condition of light or darkness into an electrical voltage change. This voltage is then used by the comparator to decide whether the relay should be activated.

6.4.3 Why is the light-dependent resistor used with a fixed resistor?

The light-dependent resistor is used with a fixed resistor to form a voltage divider. A comparator cannot directly compare resistance. It compares voltage. Therefore, the resistance change of the light-dependent resistor must be converted into a voltage change.

6.4.4 What is the voltage divider equation for the sensor node?

For the arrangement in which the fixed resistor is connected from the positive supply to the sensed node, and the light-dependent resistor is connected from the sensed node to ground, the sensed voltage is calculated as:

V_{SENSE} equals V_{CC} multiplied by R_{LDR} divided by the sum of R_{FIXED} and R_{LDR} .

In symbolic form:

$$V_{SENSE} = V_{CC} \times R_{LDR} \div (R_{FIXED} + R_{LDR})$$

6.4.5 What happens to the sensed voltage when darkness increases?

In the selected arrangement, the resistance of the light-dependent resistor increases in darkness. Since the light-dependent resistor is the lower part of the divider, the sensed voltage increases when darkness increases.

6.4.6 What condition should activate the relay in this design?

The relay should activate when the sensed voltage becomes greater than the reference voltage.

The condition is:

$$V_{SENSE} > V_{REF}$$

When this condition is true, the circuit interprets the environment as dark enough to activate the relay.

6.4.7 Why must the threshold be adjustable?

The threshold must be adjustable because the exact light level required for activation may vary according to the environment, the light-dependent resistor model, and the desired sensitivity of the circuit. A potentiometer allows the reference voltage to be tuned until the relay switches at the intended darkness level.

6.5 Comparator Questions

6.5.1 What is the function of the comparator in this project?

The comparator compares two voltages: the sensed voltage from the light-dependent resistor divider and the reference voltage from the threshold circuit. Its output changes state according to which input voltage is greater.

6.5.2 Why is a comparator used instead of directly connecting the sensor to the relay driver?

A comparator is used because the sensor voltage changes gradually, while the relay driver requires a clear switching decision. The comparator converts the gradual analog change into a definite control state that can command the driver stage.

6.5.3 Which two voltages are compared by the comparator?

The comparator compares the sensed voltage and the reference voltage. The sensed voltage comes from the light-dependent resistor divider. The reference voltage comes from the adjustable potentiometer.

6.5.4 What is the dark-activation condition for the comparator?

For the selected wiring, dark activation occurs when:

$$V_{SENSE} > V_{REF}$$

The sensed voltage is connected to the non-inverting input, and the reference voltage is connected to the inverting input. When the sensed voltage exceeds the reference

voltage, the comparator output moves toward the active state required by the driver circuit.

6.5.5 Why does the LM393 output need a pull-up resistor?

The LM393 output needs a pull-up resistor because its output stage cannot produce a valid high voltage by itself. The pull-up resistor connects the output node to the positive supply through a resistance, allowing the output to become high when the internal output transistor is not pulling it low.

6.5.6 What happens if the pull-up resistor is missing?

If the pull-up resistor is missing, the comparator output may float. A floating output has no reliable high state. This can cause the relay to remain off, switch unpredictably, chatter near the threshold, or fail to drive the transistor or metal oxide semiconductor field effect transistor correctly.

6.5.7 Why should the comparator output not power the relay coil directly?

The comparator output should not power the relay coil directly because the relay coil requires significantly more current than a comparator output is intended to provide. The comparator should only provide a control signal. The relay coil current must be supplied through a driver stage.

6.6 Relay Driver Questions

6.6.1 What is the function of the relay driver stage?

The relay driver stage allows a small comparator output signal to control the larger current required by the relay coil. It acts as an electronic switch between the relay coil and ground.

6.6.2 Why is a transistor or metal oxide semiconductor field effect transistor required?

A transistor or metal oxide semiconductor field effect transistor is required because the relay coil current is too large for the comparator output to supply directly. The driver device handles the relay coil current, while the comparator only controls the driver input.

6.6.3 What is meant by low-side switching?

Low-side switching means that the relay coil is connected to the positive supply on one side, while the switching device is placed between the other side of the coil and ground. When the switching device turns on, it completes the path from the coil to ground and allows current to flow.

6.6.4 What is the correct low-side driver connection?

The relay coil high side is connected to the positive 5 V supply. The relay coil low side is connected to the drain of the metal oxide semiconductor field effect transistor. The source is connected to ground. The gate receives the comparator output through a small resistor. A large resistor from gate to ground keeps the device off when no valid control signal is present.

6.6.5 Why is a gate resistor used?

A gate resistor limits sudden charging current into the gate capacitance and helps reduce switching disturbance. In a simulation and in a physical circuit, it also gives a cleaner and more controlled gate connection.

6.6.6 Why is a gate pull-down resistor used?

A gate pull-down resistor keeps the gate at ground potential when the comparator output is disconnected, floating, or inactive. Without this resistor, the gate may hold charge and the relay may turn on unexpectedly.

6.6.7 What is the relay coil current equation?

The relay coil current is calculated from the coil voltage and coil resistance:

$$I_{COIL} = V_{COIL} \div R_{COIL}$$

If the relay coil voltage is 5 V and the relay coil resistance is 70 Ω , then:

$$I_{COIL} = 5V \div 70\Omega$$

$$I_{COIL} = 0.071A$$

$$I_{COIL} = 71mA$$

6.6.8 What does this current value prove?

This current value proves that the relay coil requires a current that is much larger than the current normally expected from a comparator output. Therefore, the relay driver stage is not optional. It is necessary for correct operation.

6.7 Flyback Diode Questions

6.7.1 What is the function of the flyback diode?

The flyback diode protects the driver device and the control circuit from the reverse voltage produced by the relay coil when the relay is turned off. It provides a safe path for the coil current as the magnetic field collapses.

6.7.2 Why does the relay coil produce a voltage spike?

The relay coil is an inductor. An inductor resists sudden changes in current. When the driver device turns off, the coil current cannot instantly become zero. The coil therefore produces a voltage of opposite polarity in an attempt to keep current flowing. This voltage can become high enough to damage the driver or disturb the simulation.

6.7.3 How should the flyback diode be connected?

The flyback diode must be connected across the relay coil. The cathode, which is the banded side of the diode, must connect to the positive 5 V side of the coil. The anode must connect to the low side of the coil, which is also connected to the driver drain.

6.7.4 Why is the diode reverse-biased during normal operation?

The diode is reverse-biased during normal operation so that it does not short the relay coil supply. It must remain off while the relay coil is being energized. It conducts only when the coil voltage reverses during turn-off.

6.7.5 What happens if the flyback diode is missing?

If the flyback diode is missing, the relay coil generates a high reverse voltage when the driver turns off. This may cause a simulation crash, unstable switching, or damage to the driver device in a real circuit.

6.7.6 What happens if the diode is connected in the wrong direction?

If the diode is connected in the wrong direction, it may conduct during normal operation and create a near short circuit across the relay coil supply. This can prevent the relay from operating correctly and may damage components.

6.7.7 Why is the flyback diode essential rather than optional?

The flyback diode is essential because the relay coil is inductive by nature. Every inductive coil stores magnetic energy while current flows. When current is interrupted, that stored energy must go somewhere. The diode provides the controlled discharge path required for reliable switching.

6.8 Calculation Questions

6.8.1 How is the sensed voltage calculated?

The sensed voltage is calculated using the voltage divider equation:

$$V_{SENSE} = V_{CC} \times R_{LDR} \div (R_{FIXED} + R_{LDR})$$

This applies when the fixed resistor is connected from the positive supply to the sensed node, and the light-dependent resistor is connected from the sensed node to ground.

6.8.2 Calculate the sensed voltage when VCC is 5 V, RFIXED is 10 kΩ, and RLDR is 2 kΩ.

$$V_{SENSE} = 5V \times 2k\Omega \div (10k\Omega + 2k\Omega)$$

$$V_{SENSE} = 5V \times 2k\Omega \div 12k\Omega$$

$$V_{SENSE} = 5V \times 0.1667$$

$$V_{SENSE} = 0.8335V$$

The sensed voltage is approximately 0.83 V.

6.8.3 Calculate the sensed voltage when VCC is 5 V, RFIXED is 10 kΩ, and RLDR is 50 kΩ.

$$V_{SENSE} = 5V \times 50k\Omega \div (10k\Omega + 50k\Omega)$$

$$V_{SENSE} = 5V \times 50k\Omega \div 60k\Omega$$

$$V_{SENSE} = 5V \times 0.8333$$

$$V_{SENSE} = 4.1665V$$

The sensed voltage is approximately 4.17 V.

6.8.4 What do these two voltage calculations prove?

These calculations prove that the sensed voltage is low in bright light and high in darkness for the selected divider orientation. Therefore, the comparator can use this voltage to distinguish between bright and dark conditions.

6.8.5 If the reference voltage is 2.5 V, which condition activates the relay?

The relay activates when the sensed voltage is greater than 2.5 V.

The activation condition is:

$$V_{SENSE} > 2.5V$$

In the bright condition, V_{SENSE} is approximately 0.83 V, so the relay remains off. In the dark condition, V_{SENSE} is approximately 4.17 V, so the relay turns on.

6.8.6 How is relay coil current calculated?

Relay coil current is calculated using Ohm's law:

$$I_{COIL} = V_{COIL} \div R_{COIL}$$

If the relay coil voltage is 5 V and the coil resistance is 70 Ω , then:

$$I_{COIL} = 5V \div 70\Omega$$

$$I_{COIL} = 0.071A$$

$$I_{COIL} = 71mA$$

6.8.7 How is relay coil power calculated?

Relay coil power is calculated using:

$$P_{COIL} = V_{COIL} \times I_{COIL}$$

$$\text{Using } V_{COIL} = 5V \text{ and } I_{COIL} = 0.071A :$$

$$P_{COIL} = 5V \times 0.071A$$

$$P_{COIL} = 0.355W$$

The coil power is approximately 0.36 W.

6.8.8 How is load current calculated?

Load current is calculated using the power equation:

$$I_{LOAD} = P_{LOAD} \div V_{LOAD}$$

If the lamp power is 60 W and the supply voltage is 220 V, then:

$$I_{LOAD} = 60W \div 220V$$

$$I_{LOAD} = 0.273A$$

The load current is approximately 0.27 A.

6.8.9 Why must relay contact rating be checked?

Relay contact rating must be checked because the contacts must safely carry and interrupt the load current at the load voltage. A relay coil voltage of 5 V does not automatically mean the contacts are suitable for a 220 V load. Coil rating and contact rating are different specifications.

6.8.10 What calculation proves that a driver stage is necessary?

The relay coil current calculation proves that the driver stage is necessary. A relay coil current of about 71 mA is too high for the comparator output to supply directly. Therefore, a transistor or metal oxide semiconductor field effect transistor must be used to switch the coil current.

6.9 Proteus Scheme Questions

6.9.1 Which components are required in the Proteus schematic?

The required components are a 5 V direct current supply, ground, light-dependent resistor, fixed resistor, potentiometer, comparator, pull-up resistor, relay driver device, gate resistor, gate pull-down resistor, 5 V relay, flyback diode, alternating current source, fuse, and lamp.

6.9.2 How should the light-dependent resistor divider be wired?

The positive 5 V supply should connect to the fixed resistor. The other side of the fixed resistor should connect to the sensed node. The light-dependent resistor should connect from the sensed node to ground. The sensed node should connect to the comparator input.

6.9.3 How should the reference voltage be wired?

The top terminal of the potentiometer should connect to the positive 5 V supply. The bottom terminal should connect to ground. The wiper terminal should provide the reference voltage to the comparator.

6.9.4 How should the comparator be wired?

The comparator supply pin should connect to the positive 5 V supply. The comparator ground pin should connect to ground. The sensed voltage should connect to the non-inverting input. The reference voltage should connect to the inverting input. The output should connect to the pull-up resistor and to the relay driver input path.

6.9.5 How should the pull-up resistor be wired?

The pull-up resistor should connect from the positive 5 V supply to the comparator output node. This creates a valid high output state when the comparator output transistor is not pulling the node low.

6.9.6 How should the relay coil be wired?

One side of the relay coil should connect to the positive 5 V supply. The other side of the relay coil should connect to the drain of the low-side driver device. The source of the driver device should connect to ground.

6.9.7 How should the flyback diode be wired in the Proteus schematic?

The diode must be placed directly across the relay coil. The cathode must connect to the positive 5 V side of the coil. The anode must connect to the low side of the coil at the driver drain.

6.9.8 How should the lamp be wired through the relay contacts?

The live line of the alternating current source should connect to the fuse. The fuse should connect to the relay common contact. The normally open contact should connect to one terminal of the lamp. The other terminal of the lamp should connect to the neutral line of the alternating current source.

6.9.9 What is the most important scheme rule?

The most important scheme rule is that the low-voltage control side and the high-voltage load side must not be directly mixed. The relay coil belongs to the 5 V side. The relay contacts belong to the load side.

6.10 Simulation Questions

6.10.1 What is the purpose of simulation in this project?

The purpose of simulation is to verify that the circuit behaves according to the intended design before any physical implementation is considered. Simulation allows the relay switching, comparator threshold, sensor response, load activation, and protection behavior to be observed.

6.10.2 What should happen in the bright-light condition?

In the bright-light condition, the resistance of the light-dependent resistor is low. In the selected divider arrangement, the sensed voltage is low. If the sensed voltage is below the reference voltage, the relay driver remains inactive, the relay coil is not energized, and the lamp remains off.

6.10.3 What should happen in the dark condition?

In the dark condition, the resistance of the light-dependent resistor is high. In the selected divider arrangement, the sensed voltage rises. If the sensed voltage becomes greater than the reference voltage, the comparator activates the relay driver, the relay coil energizes, the normally open contact closes, and the lamp turns on.

6.10.4 Which nodes should be observed during simulation?

The important nodes are the sensed voltage node, the reference voltage node, the comparator output node, the relay coil low-side node, the relay contact path, and the lamp voltage. Observing these nodes confirms whether each stage of the circuit is working correctly.

6.10.5 What confirms that the relay is working correctly?

The relay is working correctly if the coil energizes when the dark condition is reached, the contact changes state, and the lamp turns on through the normally open contact path.

6.10.6 What confirms that the comparator threshold is working correctly?

The comparator threshold is working correctly if the output changes state when the sensed voltage crosses the reference voltage. The transition should occur near the selected threshold value.

6.10.7 What confirms that the flyback diode is working correctly?

The flyback diode is working correctly if the relay turns off without producing simulation instability, excessive voltage spikes, or driver stress. The circuit should remain stable when the relay coil is de-energized.

6.10.8 What are the acceptance criteria for the simulation?

The simulation is accepted when the lamp remains off in bright light, turns on in darkness, the relay coil energizes only when required, the comparator output changes at the threshold, the load circuit is switched through the relay contacts, and the flyback diode prevents turn-off instability.

6.11 Troubleshooting Questions

6.11.1 What should be checked first if the relay does not click?

The relay coil supply should be checked first. One side of the coil must receive 5 V, and the other side must be connected to the driver device. The driver device must also have a valid control signal at its input.

6.11.2 What should be checked if the lamp does not turn on even though the relay clicks?

The relay contact wiring should be checked. The lamp must be connected through the common and normally open contacts if it is intended to turn on only when the relay energizes.

6.11.3 What should be checked if the lamp is always on?

The relay contact selection should be checked. If the common and normally closed contacts are used, the lamp may be on when the relay is inactive. The correct connection for dark-activated turn-on behavior is common to normally open.

6.11.4 What should be checked if the relay is always on?

The comparator input polarity and reference voltage should be checked. If the sensed voltage and reference voltage inputs are reversed, or if the reference threshold is set incorrectly, the comparator may keep the relay driver active continuously.

6.11.5 What should be checked if the relay chatters?

The threshold setting should be checked first. Relay chatter can occur when the sensed voltage is very close to the reference voltage. It may also occur due to a floating comparator output, missing pull-up resistor, unstable supply, or lack of hysteresis.

6.11.6 What should be checked if the comparator output does not change?

The sensed voltage, reference voltage, comparator supply, comparator ground connection, and pull-up resistor should be checked. The comparator cannot operate correctly if any of these required connections are missing.

6.11.7 What should be checked if the simulation crashes when the relay turns off?

The flyback diode should be checked. The diode may be missing or connected with the wrong polarity. The correct diode connection is cathode to the positive 5 V side of the relay coil and anode to the low side of the relay coil.

6.11.8 What should be checked if the relay driver becomes active unexpectedly?

The gate pull-down resistor should be checked. If the driver input is floating, the relay may turn on without a valid comparator command. A pull-down resistor ensures that the driver remains off when no valid high signal is present.

6.11.9 What is the correct troubleshooting order?

The correct troubleshooting order is to verify the power supply, verify the sensor voltage, verify the reference voltage, verify the comparator output, verify the driver input, verify the relay coil voltage, verify the flyback diode polarity, verify the contact wiring, and finally verify the load circuit.

6.12 Safety and Isolation Questions

6.12.1 Why is safety important in this project?

Safety is important because the project involves the concept of switching a higher-voltage load using a low-voltage control circuit. Even when the work is performed in simulation, the report must show that the designer understands the difference between a safe simulation model and a safe physical circuit.

6.12.2 What is the main isolation rule?

The main isolation rule is that the low-voltage direct current control side and the higher-voltage alternating current load side must not be directly connected. The relay coil is part of the control side. The relay contacts are part of the load side.

6.12.3 Why must the relay contact rating be checked?

The relay contact rating must be checked because the contacts must safely switch the required load voltage and current. A relay with a 5 V coil is not automatically safe for every high-voltage load. Coil rating and contact rating are separate.

6.12.4 Why should a fuse be placed on the load side?

A fuse should be placed on the load side to protect the load circuit against excessive current. It limits damage if a fault or short circuit occurs in the load path.

6.12.5 Why is physical spacing important in real hardware?

Physical spacing is important because high-voltage conductors must be separated from low-voltage control traces and components. Insufficient spacing can lead to arcing, breakdown, shock risk, and equipment damage.

6.12.6 Why should this circuit be treated differently in simulation and in real hardware?

In simulation, components are idealized and dangerous physical effects may not appear fully. In real hardware, high voltage requires proper insulation, enclosure, rated components, fuse protection, mechanical strain relief, and qualified safety practice.

6.12.7 What is the most serious wiring mistake?

The most serious wiring mistake is directly connecting the high-voltage load side to the low-voltage control ground. This destroys the separation between the two domains and creates a dangerous circuit.

6.13 Viva and Oral Defense Questions

6.13.1 Explain the whole project in one formal answer.

This project is a light-activated relay power driver. The light-dependent resistor senses the light level and forms a voltage divider with a fixed resistor. The sensed voltage is compared with an adjustable reference voltage by a comparator. When the sensed voltage indicates darkness, the comparator output activates a relay driver. The relay driver energizes the relay coil. The relay contact then closes the load circuit and turns on the lamp. A flyback diode across the relay coil protects the driver from the reverse voltage produced when the coil is turned off.

6.13.2 Why is the relay not driven directly by the light-dependent resistor?

The relay is not driven directly by the light-dependent resistor because the light-dependent resistor is a sensing component, not a power driver. It changes resistance according to light intensity, but it does not provide a stable switching threshold or the current required by the relay coil.

6.13.3 Why is the comparator necessary?

The comparator is necessary because it converts the analog sensor voltage into a clear switching decision. Without the comparator, the relay driver would receive an undefined or gradual signal that may cause unstable switching.

6.13.4 Why is the relay driver necessary?

The relay driver is necessary because the relay coil needs more current than the comparator output should provide. The driver stage acts as the current switch for the relay coil.

6.13.5 Why is the flyback diode necessary?

The flyback diode is necessary because the relay coil stores magnetic energy while energized. When the driver turns off, the coil releases this stored energy as a reverse voltage. The diode gives this current a controlled path and protects the driver.

6.13.6 How does the circuit know when it is dark?

The circuit identifies darkness by measuring the sensed voltage produced by the light-dependent resistor divider. In the selected divider arrangement, darkness increases the resistance of the light-dependent resistor and raises the sensed voltage. When the sensed voltage becomes greater than the reference voltage, the comparator interprets this as darkness.

6.13.7 What proves that the circuit is working correctly in Proteus?

The circuit is working correctly if the lamp remains off in bright light, turns on in darkness, the relay clicks or changes contact state, the comparator output changes at the selected threshold, and the relay coil switches without simulation instability.

6.13.8 What is the strongest technical weakness of an incomplete version of this project?

The strongest technical weakness of an incomplete version is treating the relay and diode as the only required components. A correct design must also include a sensor divider, reference voltage, comparator, pull-up resistor, relay driver, flyback diode, relay contacts, load wiring, and safety separation.

6.14 Final Answer Key

The main purpose of the project is to control a higher-power load using a low-power light-sensing circuit.

The light-dependent resistor detects the light condition by changing resistance according to illumination.

The fixed resistor and light-dependent resistor form a voltage divider that converts resistance change into voltage change.

The sensed voltage is calculated as:

$$V_{SENSE} = V_{CC} \times R_{LDR} \div (R_{FIXED} + R_{LDR})$$

$$\text{For } V_{CC} = 5V, R_{FIXED} = 10k\Omega, \text{ and } R_{LDR} = 2k\Omega :$$

$$V_{SENSE} = 0.83V$$

For $V_{CC} = 5V$, $R_{FIXED} = 10k\Omega$, and $R_{LDR} = 50k\Omega$:

$$V_{SENSE} = 4.17V$$

The reference voltage is set by the potentiometer.

The relay activates when:

$$V_{SENSE} > V_{REF}$$

If $V_{REF} = 2.5V$, the relay remains off at 0.83 V and turns on at 4.17 V.

The comparator compares the sensed voltage and the reference voltage.

The LM393 output requires a pull-up resistor because it cannot produce a valid high output state by itself.

The relay driver is required because the relay coil current is greater than the current that the comparator output should supply.

The relay coil current is calculated as:

$$I_{COIL} = V_{COIL} \div R_{COIL}$$

For $V_{COIL} = 5V$ and $R_{COIL} = 70\Omega$:

$$I_{COIL} = 71mA$$

The relay coil power is calculated as:

$$P_{COIL} = V_{COIL} \times I_{COIL}$$

For $V_{COIL} = 5V$ and $I_{COIL} = 0.071A$:

$$P_{COIL} = 0.355W$$

The load current is calculated as:

$$I_{LOAD} = P_{LOAD} \div V_{LOAD}$$

For $P_{LOAD} = 60W$ and $V_{LOAD} = 220V$:

$$I_{LOAD} = 0.273A$$

The relay coil side belongs to the low-voltage control circuit.

The relay contact side belongs to the load circuit.

The common and normally open contacts are used when the lamp must turn on only after relay activation.

The flyback diode must be connected across the relay coil.

The diode cathode must connect to the positive 5 V side of the coil.

The diode anode must connect to the low side of the coil at the driver drain.

The flyback diode prevents the reverse voltage spike produced when the relay coil is turned off.

A missing flyback diode may cause driver stress, unstable switching, or simulation failure.

The most important safety rule is that the control side and load side must not be directly mixed.

The simulation is correct when the lamp is off in bright light and on in darkness.

The troubleshooting process must begin with supply verification and proceed through sensor voltage, reference voltage, comparator output, driver operation, relay coil voltage, diode polarity, relay contacts, and load wiring.

6.15 Chapter Summary

This chapter has presented the complete question structure for evaluating the light-activated relay power driver project. The questions verify conceptual understanding, component function, mathematical correctness, schematic accuracy, simulation behavior, troubleshooting ability, and safety awareness.

The chapter confirms that the project is not only a relay switching exercise. It is a complete control and power-interface system. The light-dependent resistor performs sensing. The voltage divider creates a measurable signal. The comparator creates the switching decision. The pull-up resistor makes the comparator output electrically valid. The driver stage supplies the relay coil current. The relay provides contact-based switching. The flyback diode protects the driver during coil turn-off. The load side is switched through the relay contacts and must remain separated from the low-voltage control side.

A student who can answer these questions correctly can defend the project at both schematic and functional levels. The correct explanation must always connect each component to the whole system. The final understanding is that reliable relay control requires sensing, comparison, current driving, electromagnetic switching, inductive protection, contact-side load control, and safety separation.